Evolving a One-Piece Titanium Karambit for Laser Powder-Bed Fusion under Sourced Structural, Ergonomic and Drainage Requirements

Jepson Taylor

***Abstract*—We report the best individual of the G5 v6.0.0 cohort of record (`kv4i1404_g00581_29ea3c73`), a one-piece Ti-6Al-4V karambit designed for laser powder-bed fusion (L-PBF) under the sourced requirement set RS-karambit-v5 and the continuous objective W_ERGOLIGHT_v2. Following the diagnostic audit of the preceding G3 release, G5 rebuilds the representation, material model, structural load cases, powder-drainage measurement and quality-diversity search from literature-sourced first principles. The G5.0 decoder maps 134 active phenotype genes directly in requirement units to a hollow grip and blade core filled with a sheet triply periodic minimal-surface (TPMS) lattice (≥ 0.8 mm wall), powder-removal ports (≥ 3.0 mm) placed clear of the mid-plane rib, a point-angle module whose included angle at 3 mm is penalized above 60° (60.4° on `kv4i1404_g00581_29ea3c73`), and dual-edge primary grinds (20–25° per side) with rounded-gullet serrations ($r_g \geq 0.5$ mm). Material allowables use ASTM F2924 minimum stress-relieved Ti-6Al-4V properties derated to $80\,^\circ$C service temperature, a single-application thin-wall knockdown law per face pair, and surface-class FKM high-cycle fatigue. Structural adequacy is evaluated across 16 load cases combining whole-blade sandwich-plate finite elements, a 12-direction ring sweep, Rayleigh–Ritz tip buckling, all-tooth serration checks (including end teeth), and a three-case quadratic tetrahedral (TET10) solid-tip submodel (150 N axial, 10 N lateral, and 0.30 N·m local tip twist). Release v6.0.0 continues the v5.0.0 cohort with the decoder and objective unchanged, on one platform of record whose results are byte-identical across four hosts: from the 1,508 re-evaluated v5.0.0 elites, 247 warm-started islands ran 20 rounds (296,400 evaluations) with migration through a recorded pool lineage, raising the best search-grid score from 63.36 to 75.54, above the best of 4 fresh-start control islands (59.56). Re-evaluated at Tier 2 ($h = 0.2$ mm), the winning design `kv4i1404_g00581_29ea3c73` scores 74.40 under W_ERGOLIGHT_v2 (hand multiplier $E = 0.922$, worst-margin multiplier $W = 1.000$), higher than the v5.0.0 winner re-evaluated on the same platform (59.34), at a finished partial-volume mass of 88.41 g (90.29 g on the $h = 0.2$ mm watertight mesh), and meets all 16 load cases. We render the finite-element field of every load case, break the score into the points each requirement earns and the points the multipliers take, and report why a GPU pre-screen cannot pay for itself in this search. We detail the literature provenance, decoder, load models, objective, search, verification studies, evolutionary trajectory, and every remaining limitation of the computational result.**

***Index Terms*—Evolutionary design, quality diversity, MAP-Elites, additive manufacturing, Ti-6Al-4V, sheet TPMS, finite elements, knife design.**

Release v6.0.0, decoder commit `bf3e37fa` (decoder hash `c63629d4b310`). Every computed number and every figure of the design in this paper is generated from the release receipt evidence/g5/v6.0.0/release.json. Status: computational design; not built, printed or physically tested.

1 Introduction

A karambit is a curved knife with a retention ring at the pommel, held in a closed fist with the little finger through the ring. Fabricating a one-piece karambit in Ti-6Al-4V by laser powder-bed fusion (L-PBF) makes a hollow interior possible: the grip and the blade core can be built as a thin outer skin over a triply periodic minimal-surface (TPMS) lattice with ports to evacuate unmelted powder. That structural freedom also creates failure modes where an evolutionary search can exploit its own evaluation surrogates. In the preceding G3 release (v3.0.0 and the v3.0.3 diagnostic audit), six concrete gaps emerged between what the search rewarded and what the geometry would actually do:

1) **Grid-dependent mass and drainage.** A coarse $h = 0.4$ mm voxel count undercounted mass relative to the 0.2 mm print mesh, and a single-voxel air path could drain an internal cavity at 0.4 mm while closing off at 0.2 mm. Worse, the G3 winner's 2.8 mm ports straddled its 1.1 mm mid-plane rib, splitting each port into two 0.85 mm slits that fail the 1.0 mm powder-drainage threshold [38].
2) **Additive soft objective.** In W_ERGOLIGHT_v1, safety-factor ratios $\nu = \mathrm{SF}/\mathrm{SF}_{\mathrm{req}}$ earned $0.94\,\nu$ linearly below 1.0, and mass carried weight 0.370 while individual structural terms carried 0.0174–0.0700. The search rationally traded structural adequacy on the solid tip and hollow grip for grams of mass credit.
3) **Unphysical tip and spine loads.** G3 applied the full 1.82 N·m handle twist to the last 3 mm of the point about the global origin-to-point chord. Because a sharp titanium point ($w \leq 3$ mm at 2 mm from the tip) can carry only ~ 0.2–0.45 N·m, the 1.82 N·m load taught the search to evolve a blunt, spatulate point ($> 90^\circ$ included planform angle). Conversely, G3 loaded the double-edged claw with a $3,427$ N batoning impulse taken from an Olympic boxing study [59], even though a blade sharpened on both edges has no spine to strike.
4) **Material and fatigue miscalibration.** G3 used wrought-plate A-basis yield (828 MPa), applied machined-steel Marin factors with modified Goodman mean-stress correction to as-built L-PBF surfaces [22], [27], and divided the fatigue stress by the thin-wall knockdown factor a second time after the allowable already carried it.
5) **Non-converging plate windows and unchecked end teeth.** Lens-shaped fenestration windows in the G3 blade

concentrated membrane and bending stress at sharp corners so that the whole-blade plate capacity failed to converge under mesh refinement, and the serration check skipped the end tooth at each end of a serrated span.

6) **Search bloat and fixed promotion gates.** G3 carried 179 genes (including inactive genes and a small neural network for grip texture), used Halton points rather than Lloyd-relaxed centroidal Voronoi tessellation (CVT) niche centres, and promoted candidates across tiers with fixed score thresholds that drifted as the population improved.

G5 (crates/g5, release v6.0.0) is a clean-sheet rebuild under the sourced requirement set RS-karambit-v5.toml and its literature provenance document PROVENANCE-v5.md. Every requirement is either enforced by construction in the G5.0 decoder (g5_genome.rs, g5_sdf.rs) or scored continuously by W_ERGOLIGHT_v2 (objective.rs), where the worst structural margin multiplies the entire score so that neither comfort nor mass can buy back a failed load case. This paper reports the v6.0.0 cohort of record and its best individual, `kv4i1404_g00581_29ea3c73` (decoder commit `bf3e37fa`, decoder hash `c63629d4b310`).

2 Related work and requirement provenance

Every numerical threshold, material constant, and load case in RS-karambit-v5.toml carries an explicit citation key resolved in PROVENANCE-v5.md. An automated test (RequiremtSetV5) verifies that every key cited in RS-karambit-v5.toml resolves in PROVENANCE-v5.md and that the Rust loader (requirements.rs) reads every field with deny_unknown_fields.

2.1 L-PBF Ti-6Al-4V properties, thin walls and surface-class fatigue

For stress-relieved L-PBF Ti-6Al-4V (vacuum heat treatment at 700 °C for 1 h to 800 °C for 2 h on the build plate [15], [17], [18]), ASTM F2924-14(2021) [13] sets minimum room-temperature tensile properties of σ_{y0} = 825 MPa, σ_{u0} = 895 MPa, and 10 % elongation, with fracture toughness $K_{Ic} \geq 50$ MPa$\sqrt{\text{m}}$ [19]. To cover hot-vehicle storage and field service at 80 °C, MIL-HDBK-5J [14] temperature knockdowns (0.91 on yield, 0.92 on ultimate, 0.97 on modulus) give service base properties of σ_y = 750.75 MPa, σ_u = 823.4 MPa, E = 106.7 GPa, G = 41.4 GPa, and ν = 0.31 at ρ = 4.43 g/cm^3. Following NASA-STD-6030 [16] for low-demand metallic AM hardware, limit loads require a yield safety factor SF_y = 1.2 (MS ≥ 0.2) and ultimate loads require SF_u = 1.3 (MS ≥ 0.3), while the pommel ring retains SF_{ring} = 2.0.

Thin L-PBF walls lose apparent strength because sintered surface powder and melt-pool roughness produce a non-load-bearing outer skin of depth T_s on each face. Xie *et al.* [20] fit the nominal-section knockdown law

$$k_t(t) = \left(1 - (T_{s,a} + T_{s,b})/t\right)^n, \tag{1}$$

with n = 1.8 for yield and n = 2.1 for ultimate strength across 0.2–1.0 mm walls. Combining Xie *et al.* with Murchio *et al.* [21] and Barricelli *et al.* [30] on build-angle roughness gives T_s = 0.03 mm on vertical and up-skin faces, T_s = 0.10 mm on down-skin faces below 45°, and T_s = 0.0 mm on post-print ground faces where at least 0.3–0.5 mm of stock removes subsurface lack-of-fusion pores [29], [17]. Because (1) is fitted to stress computed on the nominal thickness t, material.rs evaluates stiffness and stress on the nominal section and applies $k_t(t)$ exactly once, to the material allowable.

High-cycle fatigue (10^6 cycles) of L-PBF Ti-6Al-4V is governed by surface condition rather than wrought Marin factors [22], [23], [24], [25], [26]. Johnsen *et al.* [27] and Benedetti *et al.* [22] show that modified Goodman underpredicts mean-stress sensitivity on rough AM surfaces, whereas the FKM linear mean-stress correction $\sigma_{a,\text{eq}} = \sigma_a + M\,\sigma_m$ with M = 0.6 matches experimental data, and as-built notches exhibit full theoretical stress concentration ($K_f \approx K_t$) [23], [28]. Incorporating the NASA-STD-6030 60 % AM fatigue factor [16], RS-karambit-v5 assigns fully reversed ($R = -1$) 10^6-cycle fatigue allowables of 100 MPa for thick as-built walls (≥ 1.5 mm), 60 MPa for thin or down-skin as-built walls (< 1.5 mm), and 200 MPa for ground blade surfaces.

2.2 Sheet TPMS lattices and powder-drainage physics

Experimental comparisons of architected Ti-6Al-4V cellular solids show that sheet-based TPMS lattices (Gyroid and Schwarz-D) are stiffer, stronger, and less anisotropic than skeletal strut lattices at equal relative density [33], [34], [31], [32], while cellular fatigue remains well below $0.4\,\sigma_y^*$ at 10^6 cycles [35], [36]. Accordingly, RS-karambit-v5 forbids strut TPMS, enforces a minimum sheet thickness of 0.8 mm [37], and uses the Naghavi *et al.* [31] and Timercan *et al.* [32] Gibson–Ashby scaling laws $E^*/E_s = C_E\rho^{n_E}$ and $\sigma^*/\sigma_{ys} = C_s\rho^{n_s}$ for core homogenisation. For powder evacuation, Carrozza *et al.* [38] demonstrate that while 0.3 mm walls and 1.0 mm channels can be resolved in L-PBF Ti-6Al-4V, residual powder adheres inside passages narrower than 1.0 mm. Together with industrial depowdering rules [16], [39], RS-karambit-v5 requires at least two ports of diameter ≥ 3.0 mm per cavity and classifies an internal void as drained only if a passage of width ≥ 1.0 mm connects it to the exterior.

2.3 Human hand biomechanics, anthropometry and knife loads

Ergonomic bands and hand-applied limit loads are drawn from normative anthropometry and biomechanics [47], [48], [41], [42], [43]. Dodds *et al.* [40] establish a 95th-percentile male grip squeeze of 661 N, while the NASA Human Integration Design Handbook (HIDH) [41] sets a maximum crew grip of 783 N (800 N limit squeeze in RS-karambit-v5) and one-hand axial pull of 449–587 N (600 N limit pull). Seo and Armstrong [45] show that total normal contact force around a cylindrical handle integrates to ≈ 2.3× the grip squeeze force. In occupational knife cutting, Tirloni *et al.* [49] and McGorry *et al.* [58] measure mean working grip forces of 53–55 N and peak working forces of 130–131 N. Handle comfort and torque transmission peak for grip perimeters of 84–100 mm (floor 78.5 mm) [44] and elliptical cross-sections with height-to-thickness ratio $H/T \in [1.5, 1.8]$ [46], with a minimum grip length of 69 mm (three 95th-percentile male finger breadths of 23 mm [48]). Hand-contact pressure limits follow Fransson-Hall and Kilbom [52] (104 kPa all-day acceptable pressure) and Johansson *et al.* [53] / Aldien *et al.* [54] (188 kPa finger

discomfort, 560 kPa finger pain). Cochran and Riley [50] show that guard slip resistance plateaus at a guard height of 15.2 mm (178 N slip force; 300 N limit push in RS-karambit-v5).

For blade bending and torsion, Yoshii *et al.* [56] and MIL-STD-1472H [42] bound human wrist flexion/extension torque to ≈ 12 N·m, matching cutlery and survival-knife bend standards that apply lateral loads back from the point [62], [63]; RS-karambit-v5 sets a 12 N·m limit root moment and an 18 N·m ultimate root moment applied 5 mm behind the point. Forearm supination torque reaches 16.2 N·m in maximum exertion [55] and 4.7–7.0 N·m in repetitive wrist-twist tasks [57], [60], with palmar skin friction on metal bounded by $\mu \approx 0.30$ dry and 0.15 wet [51]. At the point, EN 388 puncture level 4 (150 N axial) [66] exceeds human skin and clothing stab penetration forces (8–60 N) [67], [68], while Hainsworth *et al.* [69] show that tip radius (0.10–0.25 mm) and planform angle govern penetration. Primary edge angles are raised from 17.5–22.5° (G3) to 20–25° per side (40–50° inclusive) because Ti-6Al-4V requires $\tan\theta \approx 1.4$–$1.5\times$ that of hardened steel for comparable edge-rolling resistance [70], [71], [72], [61], paired with medium rounded-gullet serrations (4–6 mm pitch, 0.8–1.5 mm depth, gullet radius ≥ 0.5 mm) [73], [11].

2.4 Quality-diversity optimization and multi-fidelity mechanics

MAP-Elites [1] illuminates a behaviour space by keeping the best individual in each phenotypic niche. Vassiliades *et al.* [2] introduced centroidal Voronoi tessellations (CVT) via Lloyd's algorithm [4] to partition high-dimensional descriptor spaces uniformly, together with directional variation (Iso+LineDD) [3] seeded from low-discrepancy scrambled Sobol sequences [5], [6]. In structural evaluation, discrete Kirchhoff triangular (DKT) plate elements [9] combined with constant-strain membrane triangles model curved thin shells, marching cubes [8] and marching tetrahedra [7] extract two-manifold surfaces from implicit fields, and Rayleigh–Ritz energy methods [12] and mesh refinement studies [10] verify elastic stability and discretization error.

3 The G5.0 design space

3.1 Genome and decoder construction

A G5.0 genome is a vector of 134 active genes $\mathbf{u} \in [0,1]^{134}$ (g5_genome.rs), down from 179 in G3.0 after retiring inactive genes, the neural texture network, cusped serration parameters, and blade fenestration windows. G5Genome::decode maps $\mathbf{u}$ to named physical parameters, and g5_sdf.rs builds a signed distance field (SDF) of the finished knife:

- **Ring and grip in requirement units.** The pommel ring has a fixed 24.9 mm bore and an 8.0 mm lip width (owne r-ring). Rather than emerging indirectly from independent width and thickness offsets, the grip cross-section at the palm swell is parameterized directly by its perimeter (gr ip_perim ∈ [84, 100] mm) and its height-to-thickness ratio (grip_hot ∈ [1.2, 1.5] before dorsal/belly profiling, targeting 1.5–1.8 total section H/T), matching the decoded superellipse within 0.2 %. The integral guard is hung from the scaled grip heel so that scaling the grip never buries the guard lobe.
- **Claw blade and sharp point wedge.** The curved claw blade has straight-line reach capped at 85.725 mm (3.375 in) and sharpened arc capped at 101.6 mm (4.0 in) per edge (owner-reach). Over the last point_len ∈ [10, 18] mm before the tip, a dedicated planform point wedge caps the half-width to $r\tan(\theta_{\text{point}}/2)$ with $\theta_{\text{point}} \in [22°, 35°]$, preventing the spine thickness profile from blunting the point in plan view (owner-point). The tip reserve $\Delta L \in [6, 16]$ mm keeps the distal piercing zone free of serrations and internal hollows.
- **Sheet TPMS interior and rib-clear drainage ports.** Inside an outer skin of 0.8–1.2 mm (1.0 mm minimum on load-bearing walls), the hollow grip and blade core are filled with a sheet-TPMS lattice $|\phi_{\text{TPMS}}| - t_{\text{sheet}}/2 \le 0$ blending Gyroid and Schwarz-D surfaces at sheet wall thickness $t_{\text{sheet}} \in [0.8, 1.1]$ mm [31], [33]. Powder-evacuation ports of diameter port_d ∈ [3.0, 4.0] mm (≥ 2 per cavity) are placed at $z = \pm 0.25\, t_y$, clear of the $z = 0$ mid-plane stiffening rib, and cut one full unit cell deep ($t_{\text{skin}} + L_{\text{cell}}$) into every hollow zone so that no rib bisects a port into sub-millimetre slits.

The frozen decoder is identified by the SHA-256 digest `c63629d4b310`, computed by g5_decoder_sha256 (g5_qd.rs) over every decoder, material, load, objective, solid-tip and shared finite-element source file, RS-karambit-v5.toml, and Cargo.lock.

3.2 Edge and serration construction rules

Both the concave inner edge (edge A) and the convex spine edge (edge B) follow the five EDGE-1..5 rules: a single full-height primary grind per edge (EDGE-1), measured normal to the actual serrated edge contour (EDGE-2), with per-side angle $\theta \in [20°, 25°]$ and finished apex flat $t_{\text{apex}} \in [0.05, 0.12]$ mm (EDGE-3), carried unbroken to the point except for a 5–12 % plunge ramp by the guard (EDGE-4), and checked across all steady-grind stations at 0, 0.4 and 2.0 mm behind the apex (EDGE-5). Because the decoder represents the finished knife, the L-PBF build adds a 0.5 mm finishing allowance per ground face [29], [30] that is removed when the primary bevels and serrations are ground after stress relief.

Both edges carry medium serrations (owner-serration) over a span starting 16–24 % along the blade and ending at least $\max(\Delta L + 3, 18)$ mm before the point: pitch $p \in [4.0, 6.0]$ mm, depth $d \in [0.8, 1.5]$ mm with $d \le 0.25\, p$, at least three full-depth scallops per edge, and a rounded gullet profile whose bottom radius of curvature satisfies $r_g \ge p/(2\pi) \ge 0.5$ mm by construction [73].

4 Structural load cases and multi-fidelity models

loads.rs and tip_tet10.rs evaluate all 16 structural safety-factor margins defined in RS-karambit-v5.toml. Every limit load is compared against the governing wall's 80 °C, thin-wall-derated allowable (material.rs) with $\text{SF}_{\text{req}} = 1.2$ on yield, 1.3 on ultimate, 2.0 on the owner's ring loads, 1.0 on surface-class FKM fatigue, 1.2 on per-case blade-before-hand yield sequence, and $\lambda_{\text{req}} = 1.5$ on tip buckling.

4.1 Tier 0: integrated curvilinear blade and grip sections

At Tier 0, Midline::of traces the smoothed blade midline from the point ($s = 0$) to the guard heel, and blade_section integrates the exact decoded thickness law across the planform normal to the midline every 1.5 mm, accounting for the homogenised sheet-TPMS core stiffness where the blade is hollow. An external force $\mathbf{F}$ applied at $\mathbf{q}$ is resolved at each station into axial tension N, through-thickness bending M_b, in-plane bending M_{ip}, and Saint-Venant open-section torsion T, superposed conservatively as

$$\sigma_{\mathrm{n}} = |N|/A + |M_b|\, t_{\max}/(2I_{\mathrm{minor}}) + |M_{\mathrm{ip}}|\, c_{\mathrm{major}}/I_{\mathrm{major}},$$
$$\sigma_{\mathrm{eq}} = \sqrt{\sigma_{\mathrm{n}}^2 + 3(|T|\, t_{\max}/J)^2}. \tag{2}$$

This section chain evaluates:

- **Lateral bend (lateral_limit, lateral_ultimate):** Out-of-plane force applied 5 mm behind the point producing a 12 N·m limit root moment (checked against yield at SF ≥ 1.2) and an 18 N·m ultimate root moment (checked against ultimate strength at SF ≥ 1.3). On a curved claw, a lateral force near the tip induces coupled bending and torsion at every proximal station.
- **Body twist (twist_limit):** 16.2 N·m maximum forearm supination torque [55] about the blade axis (SF ≥ 1.2).
- **Axial pull (pull):** 600 N pull along the grip axis reacted over the distal 20 mm of the blade [41].
- **Hollow grip, guard and ring (grip_squeeze, guard_push, ring_*):** The hollow grip wall and internal web are checked under the 800 N distributed hand squeeze (normal force sum 2.3×800 N [45]) and under the reaction wrenches of the lateral, twist (with at most 2.0 N·m palmar friction credit [46], [51]) and pull cases. The guard lobe is checked as a root section under a 300 N forward hand slip [50]. The pommel ring is solved with quadratic six-node membrane triangles (T6) under 300 N bearing (±60° arc, SF ≥ 2.0), 30 N back pull (SF ≥ 2.0), and a 12-direction in-plane sweep of 300 N every 30° (SF ≥ 1.2).
- **All-tooth serration check (teeth):** Every tooth on both edges—including both end teeth bounded by a gullet on one side and the plain edge on the other—is checked on the 2D section law under a 180 N in-plane snag (with the Inglis notch factor $K_t = 1 + 2\sqrt{h_t/\rho}$ of its rounded gullet [11]) and a 60 N lateral pry shared over three teeth (20 N per tooth).
- **Surface-class FKM fatigue (fatigue) and per-case sequence (sequence):** High-cycle fatigue (10^6 cycles) combines the 4.7 N·m reversed lateral root moment and 7.0 N·m twist duty [49], [58] on the ground blade against the 200 MPa ground allowable, and checks the hollow grip and ring under the working grip (130 N peak) and little-finger (25 N) duty with FKM mean-stress sensitivity $M = 0.6$ and $K_f = K_t$ on as-built faces [22], [27], [23]. The yield-sequence ratio $\mathrm{seq} = \min_{c\in\{\mathrm{lat,\ twist,\ pull}\}} \mathrm{SF}_{\mathrm{hand}}(c)/\mathrm{SF}_{\mathrm{blade}}(c)$ is evaluated per load case separately (per_load_case = true, target ≥ 1.2).
- **Tip buckling (buckling):** Rayleigh–Ritz two-mode eigenvalue solve [12] of the tapered distal blade cantilevered 30 mm from the point under the 150 N axial tip compression load, using wide-plate flexural rigidity $D(s) = E\,b(s)\,t(s)^3/[12(1-\nu^2)]$ integrated over 60 stations ($\lambda_{\mathrm{req}} = 1.5$).

4.2 Tier 1: whole-blade sandwich plate

At Tier 1, plate_blade_cases (loads.rs) replaces the Tier-0 section estimates for lateral_limit, lateral_ultimate, twist_limit, pull, fatigue, and sequence with a whole-blade SandwichPl ate finite-element solve (constant-strain membrane triangles plus discrete Kirchhoff bending triangles [9]) at production maximum triangle area 0.3 mm^2. Retiring G3's sharp-cornered fenestration windows and distributing the lateral patch load around 5 mm from the point, the torsional couple over the working span, and the 600 N pull over the distal 20 mm removes window-corner singularities; across the four verification meshes ($1.2, 0.6, 0.3, 0.15$ mm^2 in vv/plate_mesh.json), lateral safety factors vary non-monotonically by up to 20.3 % while twist safety factors stay within 5.6 % (Section 7).

4.3 Tier 2: three-case quadratic tetrahedral (TET10) solid tip

At Tier 2, tip_solid_cases (tip_tet10.rs) replaces the beam-section tip estimates with a 3D quadratic 10-node tetrahedral (TET10) submodel of the distal 30 mm of the blade, meshed at maximum planform triangle area 0.08 mm^2 across four thickness layers (cell-centred thickness sampling) and clamped at the $r_{\mathrm{cut}} = 30$ mm arc. Three independent linear-elastic solves are executed on the submodel under the sourced RS-karambi t-v5 tip loads:

1) **Tip axial (150 N, EN 388 puncture level 4 [66]):** In-plane compression along the local tip bisector, distributed over a load patch of radius 0.75 mm at the point; evaluated over the readout window $r \in [1.0, 10.0]$ mm (SF ≥ 1.20).
2) **Tip lateral (10 N side load):** Out-of-plane $+z$ shear distributed over a 0.75 mm patch at the point; evaluated over the readout window $r \in [1.0, 2.5]$ mm (SF ≥ 1.20).
3) **Tip twist (0.30 N·m about the local tip bisector):** Opposing $\pm z$ couple applied at $r = 3.0$ mm behind the point over a ±1.0 mm patch half-width so the discrete nodal forces sum to zero net force and an exact 300 N·mm torque about the tip bisector; evaluated over the readout window $r \in [4.5, 15.0]$ mm (SF ≥ 1.20).

Each case reports $\mathrm{SF} = \sigma_{\mathrm{allow}}/\sigma_{\mathrm{vM,max}}$ from the nodal maximum von Mises stress within its per-case readout window (re adout_mm) against the ground-blade allowable (750.75 MPa).

5 The continuous objective W_ERGOLIGHT_v2

Under the owner's rules (owner-objective), W_ERGOLIGH T_v2 (objective.rs) is strictly continuous with no hard rejection gates, while preventing comfort or low mass from compensating for a failed structural load case:

$$\mathrm{score} = 100\,E^5\,W\Big(w_{\mathrm{mass}}\,s_{\mathrm{mass}} + \sum_{i=1}^{21} w_i\,s_i\Big), \qquad E = \prod_{k=1}^{8} e_k^{\beta_k}. \tag{3}$$

Table 1 lists the weights w_i ($\sum w = 1.00$ including $w_{\mathrm{mass}} = 0.30$) and the ergonomic exponents β_k ($\sum \beta_k = 1.00$).

Table 1 | Weights w_i and ergonomic exponents β_k of W_ERGOLIGHT_v2 (RS-karambit-v5.toml)

Term s_i	w_i	Ergonomic factor e_k	β_k
Mass (130 → 25 g)	0.3000	Grip section (P, H/T)	0.20
Point geometry	0.0600	Ring (w_{lip}, crown)	0.15
AM build (void mm^3)	0.0600	Hand-contact radii	0.15
Fatigue (FKM class)	0.0400	Contact pressure	0.15
Yield sequence (≥ 1.20)	0.0400	Guard height (≥ 15 mm)	0.10
Edge bevel & apex	0.0400	Grip length (≥ 69 mm)	0.10
Lateral limit (12 N·m)	0.0350	Balance (CoM & I)	0.10
Lateral ult. (18 N·m)	0.0350	Grip surface texture	0.05
Body twist (16.2 N·m)	0.0350		
Tip axial (150 N)	0.0350	**Multipliers**	**Exp.**
Tip lateral (10 N)	0.0350	Ergonomics $E = \prod e_k^{\beta_k}$	5.0
Tip twist (0.30 N·m)	0.0350	Worst margin $W = \min(1, \mu^4)$	1.0
Grip squeeze (800 N)	0.0350		
Guard push (300 N)	0.0300		
Ring bearing (300 N)	0.0300		
Serration teeth (180/60 N)	0.0300		
Form fairness & arch	0.0250		
Ring back (30 N)	0.0200		
Ring 12-dir. (300 N)	0.0200		
Axial pull (600 N)	0.0200		
Tip buckling ($\lambda \geq 1.50$)	0.0200		
Serration depth & count	0.0200		

5.1 Quartic sub-threshold penalty and worst-margin multiplier

Each of the 16 structural terms compares its safety factor (or buckling eigenvalue) against its requirement via the ratio $v_i = \mathrm{SF}_i/\mathrm{SF}_{\mathrm{req},i}$. Instead of G3's linear ramp below 1.0, s_margin applies a quartic penalty below adequacy and a saturating exponential reserve above it:

$$s(v) = \begin{cases} 0.94\, v^4, & 0 \leq v < 1, \\ 0.94 + 0.06\,(1 - e^{-(v-1)/0.15}), & v \geq 1. \end{cases} \tag{4}$$

Thus a safety factor at 64 % of its requirement ($v = 0.64$) earns less than 0.16 of its term weight rather than 0.60. Furthermore, let $\mu = \min_{i\in\{1..16\}} v_i$ be the worst safety-factor ratio across all 16 structural cases. In our initial G5 pilot (12 islands ×600 evaluations), an additive 0.25 worst-margin term still allowed a design failing five load cases to score within 12 % of an all-passing 90 g design when E^5 multiplied a small comfort advantage. In W_ERGOLIGHT_v2, the worst margin therefore multiplies the entire score:

$$W(\mu) = \min(1,\, s(\mu)/0.94) = \min(1,\, \mu^4). \tag{5}$$

When every structural case meets its requirement ($\mu \geq 1$), $W = 1$ and the multiplier is inactive. When any single load case falls short ($\mu < 1$), $W = \mu^4$ scales down every point earned by mass, geometry, and ergonomics. A unit test in objective.rs verifies that lifting a single failing safety factor from any $v_0 \in [0.5, 0.95]$ to 1.0 outscores 10 g of mass reduction.

5.2 Geometry terms, ergonomic exponent and unmeasured quantities

The five geometry terms score distance to the RS-karambi t-v5 bands on the decoded model: primary edge angle (20–25°) and apex flat (0.05–0.12 mm); planform point included angles (≤ 75° at 1 mm, ≤ 60° at 3 mm), width at 2 mm (≤ 3.0 mm), and tip thicknesses at 0.25, 1.0 and 2.0 mm; serration depth (0.8–1.5 mm) and scallop count (≥ 3 per edge); AM build drainability $\exp(-V_{\text{sealed}}/300\ \text{mm}^3)$ at Tier 1 and above (0 at Tier 0 where voids are unmeasured); and form fairness. Raising E to the fifth power (E^5) ensures that 1 % of ergonomic shortfall costs 14.1 g of mass credit at the winning design (0.263 points per gram), exceeding the 10 g minimum in RS-karambit-v5. Quantities that require physical testing (such as ISO 8442-5 edge retention and cutting force) are not guessed by formula and earn zero credit in the score.

6 Search architecture

6.1 Lloyd-relaxed CVT archive, sparse variation and baseline

Each evolving island maintains a MAP-Elites archive [1], [2] of 1,024 niches in an eight-dimensional behaviour space (CvtMapElitesArchive): hook drop, inner turn angle, silhouette complexity, grip section H/T, finger-groove articulation, porosity fraction, surface-texture energy, and mass. Whereas G3 used raw Halton points as niche centres, G5 runs 20 iterations of deterministic Lloyd relaxation [4], [2] from Halton starts (16 fixed scrambled-Sobol samples per centre, lloyd_cvt_centroids) so that the Voronoi cells partition $[0, 1]^8$ with near-uniform volume.

In the v5.0.0 cohort every island initialized exclusively from 64 Joe–Kuo Sobol points in $[0, 1]^{134}$ [5], scrambled by an island-seeded linear matrix and digital shift [6]. Offspring are generated by selecting two parents (with probability 0.40 from the 16 highest-scoring elites, otherwise by three-way curiosity tournament) and applying sparse directional variation (reproduce_iso_line_dd): module crossover with probability 0.15, self-adaptive per-module step sizes, and a sparse mask that perturbs an average of ≈ 15 of the 134 genes per child so that localized structural or geometric improvements are not scrambled by simultaneous full-genome moves. v5.0.0 also ran random-search islands that drew every evaluation from the scrambled Sobol sequence with the same budget and tier cascade; v6.0.0 replaces them with the fresh-start control of Section 6.3.

6.2 Rank-based tier promotion and bit-exact archive verification

Fixed score gates for tier promotion either starve Tier 1 early in a run or flood it once scores rise. G5 replaces fixed thresholds with a rolling quantile promoter (RankPromotion in g5_qd.rs): a window of the last 256 Tier-0 scores (every child is promoted during the first 16) promotes a child to Tier 1 (h = 0.4 mm voxel SDF, width-aware drainage flood fill, whole-blade SandwichPlate FE, and T6 ring sweep) if and only if its Tier-0 score ranks in the top 40 % of the rolling window.

During the search cohort, Tier 1 is the highest active tier so that compute is concentrated on whole-blade plate mechanics and 3D voxel drainage across thousands of candidates. After the cohort finishes, all 282 final elites across the islands are re-evaluated at Tier 1 (h = 0.4 mm) to verify bit-exact archive reproduction, and the top 64 designs are ranked at Tier 2 on the fine print grid (h = 0.2 mm watertight marching-tetrahedra mesh census plus the three-case 0.08 mm^2 TET10 solid-tip submodel). Every persisted archive row stores its full G5Loads and AmConnectivityStats structures so that load_and_verify_pe

rsisted_dir re-evaluates W_ERGOLIGHT_v2 bit-for-bit against the recorded score.

6.3 Continuation, migration and one platform of record (v6.0.0)

v6.0.0 continues the v5.0.0 cohort with the decoder, requirement set and objective unchanged (decoder hash `c63629d4b310`). Its search runs on one platform of record: a single statically linked x86_64 binary per tool, run natively on the Linux hosts and in a linux/amd64 container under Rosetta on the macOS host. The same island gave byte-identical archives and histories on all four hosts before any run of record, which removes the v5.0.0 limit that its arm64 records were close to, but not bit-identical with, x86_64 re-evaluations.

Warm start. The 1,508 final elites of v5.0.0 (its sixteen evolving islands) were re-evaluated at Tier 1 on the platform of record and merged to the best record per niche (g5-continue reeval and merge): 131 niches, the starting pool.

Rounds and migration. Each island of a round starts from the latest pool (its records inserted after a bit-exact re-score) and runs 1,200 evaluations with its own seed. When every island of a host's round has finished, a hub merges their archives into the latest pool, keeping the best record per niche in a recorded order, and the host's next round starts from the new pool. Hosts therefore advance at their own speed; every pool records its sources, so the whole lineage is a deterministic function of the recorded inputs, and the receipt recomputes it byte for byte (Section 11).

Control. 4 fresh-start islands ran 4,800 evaluations each from scrambled Sobol samples, with the same decoder, objective and promotion and no pool or migration. They are never merged and cannot win.

Sizing on the hosts. The continuation ran one island per core complex (four cores sharing an L3 cache) on the Threadripper hosts, because an early pilot's time per evaluation rose steeply with the number of concurrent evaluations. A controlled measurement after the search does not support that choice. The same 12 Tier-1 evaluations took 5.6 s each with one evaluation per core complex and 7.7 s with four, so one per core would have given 2.89× the throughput of the setting used (fleet/sizing.json); the pilot had mixed tiers, simultaneous multithreading and other users' load in it. The search is deterministic either way, so this cost evaluations, not correctness. Freed memory is kept for reuse (glibc tunables: 2.5× fewer page faults, identical outputs).

GPUs. A profile of the Tier-1 evaluation puts 21 % of its time in the voxel distance transform, 18 % in the drainage connectivity flood fill and 20 % in the plate's skyline solve; Tier 1 is 90 % of an island's time (9.4 s against 0.41 s for Tier 0, over 200 children of the starting pool). The records of the search must stay exact CPU arithmetic, so a GPU could only shorten or skip Tier-1 work. Off-loading the voxel kernels for free bounds Tier 1 at 1.64× faster, and a perfect, free pre-screen that promoted only the true top 10 % of children at 3.09× more evaluations per hour, and the Tier-0 screen already keeps 17 of the 20 best children (Spearman 0.90). A real predictor costs time and misses designs, so the 3× bar set for adopting a GPU path is out of reach; the search stays on the CPUs and the GPUs render the figures.

7 Measurement and verification

Four measurement fixes in g5_sdf.rs and tip_tet10.rs eliminate the grid and discretization biases identified in the G3 audit:

1) **Smoothed-Heaviside partial-volume mass.** Instead of binary sign counting ($\phi \le 0$), voxel mass and moments of inertia integrate a C^1 polynomial smoothed Heaviside kernel $H_\Delta(\phi)$ over a half-width $\Delta = \frac{\sqrt{3}}{2}h$:

$$w(\phi) = \frac{1}{2} - \frac{3}{4}\frac{\phi}{\Delta} + \frac{1}{4}\Big(\frac{\phi}{\Delta}\Big)^3, \qquad |\phi| < \Delta. \tag{6}$$

   This reduces voxel-pitch mass drift across $h \in \{0.4, 0.3, 0.25, 0.2, 0.15\}$ mm to at most 0.51 % on the top designs (Table 2), well inside the 2.0 % requirement in RS-karambit-v5.
2) **Resolution-invariant morphological radii.** Hand-zone morphological closing and opening convert their physical millimetre blur radii into voxel counts via $\mathrm{round}(r_{\mathrm{mm}}/h)$, so refining h preserves the physical fillet radius instead of shrinking it.
3) **Width-aware 1.0 mm powder drainage and crevices.** enforce_am_connectivity erodes internal cavity air by the $r_{\mathrm{drain}} = 0.50$ mm half-channel radius ($\mathrm{ceil}(0.50/h)$ voxels) before flooding from the exterior and dilating back into the cavity [38]. Any undrained air is classified by connected component: a component whose air is narrower than the drainage half-width everywhere (sheet against skin) is labeled a *crevice* (crevice_mm3), whereas any component containing wider undrained air is labeled *sealed* (sealed_void_mm3). Both are printed solid in the build field ($d_{\mathrm{am}} = -h$, so undrained regions add solid titanium mass), and only sealed volume is penalized in the objective.
4) **Cell-centred solid-tip raster.** The 2D thickness raster feeding the 3D TET10 tip submodel is sampled at cell centres $(i + \frac{1}{2})h_{\mathrm{tip}}$, matching the lookup coordinates of the tetrahedral mesh builder and removing G3's half-cell shift.

Alongside the cohort receipt, g5-vv writes five deterministic verification records under evidence/g5/v6.0.0/vv/ with SHA256 SUMS: grid.json (partial-volume mass at $h = 0.4$ vs. 0.2 mm), plate_mesh.json (whole-blade SandwichPlate mesh convergence across maximum triangle areas $1.2, 0.6, 0.3, 0.15$ mm^2), tip_mesh.json (solid-tip TET10 mesh sensitivity at 0.15 and 0.08 mm^2 against the analytical section model), decoder_targets.json (decoded grip perimeter within 0.12 % of its gene, plus H/T, point angles and grip length), and checks.json (sheet-TPMS density, Ritz column buckling vs. Euler's formula, Lloyd-relaxed CVT quantization error 0.125 vs. Halton 0.143, and the 10.5 g per 1 % of E trade; Table 5).

8 Results

8.1 The continuation and its fresh-start control

Starting from the 131 niches of the re-evaluated v5.0.0 elites (best 63.36 on the $h = 0.4$ mm search grid), 247 warm-started islands ran 20 rounds of 1,200 evaluations on 2 hosts (xbabe3, xbabe4; 296,400 evaluations in all, at most

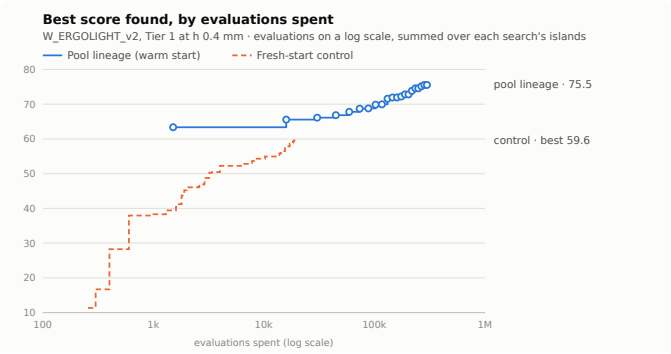


Figure 1 | Best W_ERGOLIGHT_v2 score (Tier 1, $h = 0.4$ mm) against the evaluations spent, log scale: the pool lineage from the re-evaluation of the v5.0.0 elites on (a step at each merge), and the 4 fresh-start control islands as their best against their combined evaluations (best 59.56).

14 rounds deep on one host). The pool lineage (21 pools, Fig. 1) raised the best search-grid score by 12.18 to 75.54, over 282 niches in the final pool `pools/p20`. The 4 fresh-start control islands (4,800 evaluations each, no pool) reached at best 59.56, so the lineage ended above the best fresh start. The winner of record `kv4i1404_g00581_29ea3c73` came from epoch14/xbabe4/island1404 (75.15 on the search grid, 74.40 at Tier 2 on the $h = 0.2$ mm ranking grid). Fig. 2 shows the best design of pools of the lineage from a fixed camera.

8.2 Grid check and final Tier-2 ranking

To verify grid independence, the top designs were evaluated across five voxel pitches $h \in \{0.4, 0.3, 0.25, 0.2, 0.15\}$ mm (Table 2). With smoothed-Heaviside partial-volume integration and width-aware drainage, the 0.20 mm and 0.15 mm grids agree within 0.51 %, and the search's 0.40 mm mass matches the 0.20 mm mass of record within 0.56 %. Table 3 lists the top designs from the final Tier-2 ranking (shortlist 64) at $h = 0.2$ mm and 0.08 mm^2 solid-tip mesh resolution; the top eight rows span 74.40–74.02. Re-ranking the shortlist at Tier 2 matters, because Tier 1 checks the tip with a section model and Tier 2 with the 3D TET10 solid: the search-grid leader `kv4i1406_g00009_528e4365` (75.54 at Tier 1) is at rank 2 of the Tier-2 ranking. Because the plate's lateral safety factor is not monotone in the mesh (Section 7), the winner is chosen on its plate cases re-solved at 0.6/0.3/0.15 mm^2 on the platform of record (whose production-mesh rows reproduce the ranking bit for bit): in rank order, the first design meeting every load case on every mesh. That is `kv4i1404_g00581_29ea3c73`, with worst margin $\mu = 1.094$ over the meshes (1.094 on the production mesh).

8.3 The best individual (`kv4i1404_g00581_29ea3c73`)

The winning individual, `kv4i1404_g00581_29ea3c73` (genome hash `29ea3c73be53`), scores 74.40 under W_ERGOLIGHT_v2 with weighted base 0.807, hand multiplier $E = (\prod_j e_j^{\beta_j})^{5.0} = 0.922$ ($\prod_j e_j^{\beta_j} = 0.984$), and worst-margin multiplier $W = 1.000$ (worst safety-factor ratio $\mu = 1.094$ on tip_axial). Its finished partial-volume mass of record at $h = 0.2$ mm is 88.41 g (90.29 g on the watertight print mesh; solid outer envelope 47,662 mm^3, 211.1 g). Its

Table 2 | Grid check: partial-volume mass (g) of the top designs across five voxel pitches h (mm)

Design	0.4	0.3	0.25	0.2	0.15
`kv4i1404_g00581_29ea3c73`	88.6	89.5	89.2	88.4	88.6
`kv4i1406_g00009_528e4365`	88.6	89.0	88.5	88.3	88.3
`kv4i1408_g00735_e6796cbf`	89.1	89.5	89.1	88.9	88.9
`kv4i1405_g00335_c923c290`	88.8	89.0	88.4	88.4	88.1
`kv4i1414_g00435_0e2e651b`	89.8	90.1	90.2	89.5	89.5
`kv4i1410_g00279_290be921`	89.3	89.7	89.2	89.0	88.9
`kv4i1401_g00149_412d5153`	88.9	89.4	88.7	88.6	88.5
`kv4i1409_g00336_4a44463a`	89.3	89.8	89.1	89.0	88.9
`kv4i1401_g00281_b7d5a90f`	89.6	89.9	89.7	89.1	89.1
`kv4i1409_g00504_749c79d8`	88.7	89.2	88.5	88.4	88.3
`kv4i1414_g00999_42619e63`	90.9	91.1	90.5	90.4	90.3
`kv4i1410_g00280_5831a5e3`	90.3	90.8	90.1	90.0	89.9
`kv4i1405_g00205_a7f055b6`	90.3	90.8	90.3	90.3	90.2
`kv4i1411_g00490_71efc13e`	90.3	90.7	90.0	90.0	89.8
`kv4i1310_g00526_5127c223`	90.9	91.5	90.7	90.5	91.0
`kv4i1414_g00960_db1fd3ec`	88.9	89.3	88.7	88.6	88.5

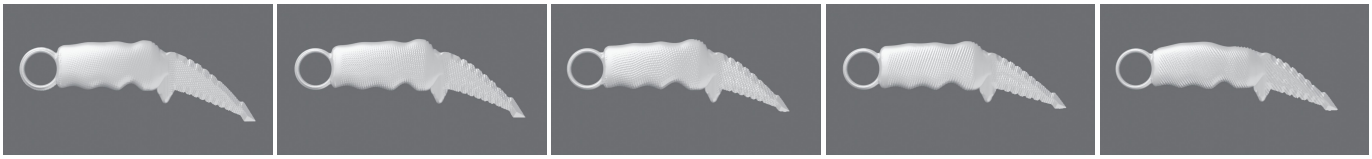
Table 3 | Final ranking: Tier 2 at $h = 0.2$ mm and 0.08 mm^2 solid-tip mesh

Rank	Design	Score	W	Mass (g)
1	`kv4i1404_g00581_29ea3c73`	74.40	1.000	88.4
2	`kv4i1406_g00009_528e4365`	74.32	1.000	88.3
3	`kv4i1408_g00735_e6796cbf`	74.18	1.000	88.9
4	`kv4i1405_g00335_c923c290`	74.06	1.000	88.4
5	`kv4i1414_g00435_0e2e651b`	74.04	1.000	89.5
6	`kv4i1410_g00279_290be921`	74.04	1.000	89.0
7	`kv4i1401_g00149_412d5153`	74.03	1.000	88.6
8	`kv4i1409_g00336_4a44463a`	74.02	1.000	89.0

grip is 75.9 mm long with a swell perimeter of 99.4 mm (84–100 mm target band) and height-to-thickness ratio 1.63 (1.5–1.8 target band); the guard is 15.8 mm high and the straight-line blade reach is 72.2 mm. Internal sealed void volume is 44.9 mm^3 (open surface crevices 199.7 mm^3), drained through 4 ports of diameter 3.93 mm. The evolved surface texture (depth 0.22 mm, pitch 2.51 mm) reaches the blade with scope 0.07 (0.014 mm deep there), so the as-decoded blade and its bevels are nearly smooth; the analytical edge cross-sections (Fig. 4) and edge metrics evaluate the bevel law restored after post-build grinding (0.5 mm finishing allowance per ground face). Fig. 3 and Fig. 4 show its geometry; Table 4 reports all 16 structural load cases and key geometric measurements against RS-karambit-v5.

8.4 Against the v5.0.0 winner on the same platform

Re-evaluated on the v6.0.0 platform of record, the v5.0.0 winner `kv4i012_g00009_dd6ffea5` scores 59.34 at Tier 2 (59.55 at Tier 1 on the search grid), with worst margin 1.014, 16 of 16 load cases met and 83.3 g; its v5.0.0 record was made on arm64. `kv4i1404_g00581_29ea3c73` scores 74.40, higher than the v5.0.0 winner, with worst margin 1.094, 16 of 16 cases met and 88.41 g (heavier than it). Its hand multiplier is $E = 0.922$ (v5.0.0: 0.740), with grip length and perimeter 75.9 and 99.4 mm (v5.0.0: 69.1 and 84.1 mm).



pool epoch00/pool 63.36, 103.6 g **warm start** | pool pools/p04 67.81, 95.9 g | pool pools/p10 71.94, 96.2 g | pool pools/p14 72.81, 90.7 g | pool pools/p20 75.54, 88.6 g **final pool**

Figure 2 | Best design of pools of the lineage: pool, W_ERGOLIGHT_v2 score and mass on the search grid. Fixed camera, lighting and material across all frames.

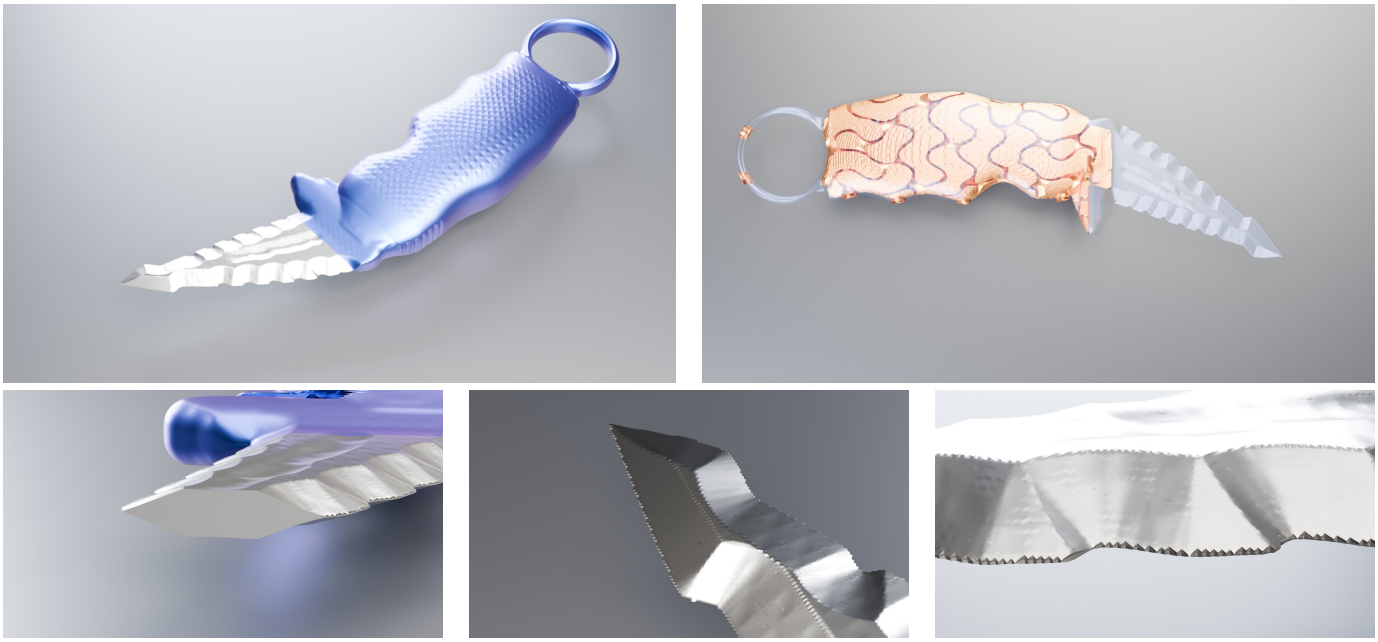


Figure 3 | `kv4i1404_g00581_29ea3c73`. Top: three-quarter view as decoded (before grinding), and an x-ray cutaway showing the internal sheet-TPMS cavities and rib-clear drainage ports in amber. Bottom: cross-section across the blade 30 mm from the point, the sharp point wedge, and the spine edge with its rounded-gullet serrations, as decoded (before grinding).

8.5 Where the score comes from

The score is $100\,EW$ base, so its 100 points split into what the design earns, what the two multipliers take and the credit its terms fall short of (Fig. 5). `kv4i1404_g00581_29ea3c73` earns 74.4: structural safety 45.3 (the 16 load-case margins), mass 11.0, cutting geometry 11.0 (edge, point and serrations) and build and form 7.1. The multipliers take 6.3 (E = 0.922, W = 1.000) and 19.3 is unmet credit (1 − base). Each term's points against the most it could earn are in Fig. 6.

9 FEM renderings of every load case

The scored solvers return safety factors and discard their fields. g5-fem-fields re-solves each of the 16 load cases of `kv4i1404_g00581_29ea3c73` on the ranking grid (h = 0.2 mm) with copies of the scored models (the whole-blade CST+DKT sandwich plate, the TET10 solid tip, the T6 ring with its twelve-direction sweep and the Rayleigh–Ritz column, now with its mode) and keeps the fields: von Mises stress on a five-band sequential scale, the governing point and the deformed shape (Figs. 7 and 8). The 28 scored values these solves produce (the safety factors, the Ritz eigenvalue, the sequence ratios and the governing points) match the receipt bit for bit on the platform of record; the tool fails if any does not.

The score models grip squeeze, guard push and teeth analytically. For those three, the figures show an independent hex8 voxel finite-element solve on the design's own additive-manufacturing voxel field instead, reported beside the score model and not used by it (Table 6).

10 Limits of the result and evidence gaps

What this result is not

It is a computational design (computational design; not built, printed or physically tested). Nothing has been printed, stress-relieved, ground or physically tested, and no safety factor reported here is a qualification for use.

- **Outcome against the targets.** The best individual `kv4i1404_g00581_29ea3c73` has worst safety-factor margin ratio μ = 1.094 (on tip_axial, multiplier W = 1.000), finished partial-volume mass 88.41 g (90.29 g on the 0.20 mm mesh, which is above the 50 g goal (frontier) and under the 130 g ceiling), and sealed internal void volume 44.9 mm^3 (printed solid in the build field, $d_{\mathrm{am}} = -h$, so the mass counts solid titanium rather than trapped powder).

Table 4 | Measurements and safety factors of the best individual `kv4i1404_g00581_29ea3c73` against RS-karambit-v5. SF on the production plate mesh; v is the margin of record, the minimum over the plate meshes 0.6/0.3/0.15 mm^2

Quantity / Load case	Value	Req.	v
Score (W_ERGOLIGHT_v2)	74.40	—	$E = 0.922$
Mass, partial vol. / mesh (g)	88.41 / 90.29	50 goal ($\leq$ 130)	$W = 1.000$
Sealed void (mm^3) / ports	44.9 / $\emptyset$3.93	0 / $\geq$ 3.0	$N = 4$
Grip L_g / perim. (mm) / H/T	75.9 / 99.4 / 1.63	$\geq$ 69 / 84–100	1.5–1.8
Guard h / CoM to choil (mm)	15.8 / 7.3	$\geq$ 15 / $\leq$ 8	—
Point angle 1 / 3 mm (°)	65.5 / 60.4	$\leq$ 75 / $\leq$ 60	$w_2 = 2.05$
Edge A / B grind (°, max 23.5)	22.3 / 23.5	20–25°	flat 0.113
Serr. depth (mm), scallops A / B	1.09 (6) / 1.30 (5)	0.8–1.5, $\geq$ 3	$r_g \geq 0.5$
Lateral limit (12 N·m, plate)	SF 2.01	$\geq$ 1.20	1.636
Lateral ult. (18 N·m, plate)	SF 1.47	$\geq$ 1.30	1.104
Body twist (16.2 N·m, plate)	SF 1.90	$\geq$ 1.20	1.583
Tip axial (150 N, TET10)	SF 1.31	$\geq$ 1.20	1.094
Tip lateral (10 N, TET10)	SF 1.86	$\geq$ 1.20	1.550
Tip twist (0.30 N·m, TET10)	SF 2.50	$\geq$ 1.20	2.083
Grip squeeze (800 N)	SF 2.32	$\geq$ 1.20	1.933
Guard slip push (300 N)	SF 26.39	$\geq$ 1.20	21.993
Ring bearing (300 N, $\pm$30°)	SF 65.39	$\geq$ 2.00	32.694
Ring back (30 N)	SF 68.46	$\geq$ 2.00	34.232
Ring in-plane sweep (300 N)	SF 3.49	$\geq$ 1.20	2.910
Axial pull (600 N, plate/grip)	SF 1.83	$\geq$ 1.20	1.399
Serration teeth (180/60 N, all)	SF 2.38	$\geq$ 1.20	1.983
Fatigue (10^6 cyc, FKM class)	SF 1.17	$\geq$ 1.00	1.171
Yield sequence (min per case)	1.91	$\geq$ 1.20	1.592
Tip buckling λ (150 N, Ritz)	11.94	$\geq$ 1.50	7.958

Table 5 | Verification studies (evidence/g5/v5.0.0/vv/, unchanged decoder, and release.json): plate refinement, solid-tip refinement and solver checks

Study	Quantity	Coarse	Med.	Fine
Plate (1.2/0.3/0.15 mm^2)	Lat. limit SF	1.69	1.68	1.84
Plate (1.2/0.3/0.15 mm^2)	Twist limit SF	1.53	1.53	1.54
Tip TET10 (sec/0.15/0.08)	Axial SF	1.37	1.21	1.27
Tip TET10 (sec/0.15/0.08)	Lateral SF	1.02	1.42	1.26
Tip TET10 (sec/0.15/0.08)	Twist SF	0.73	1.32	1.54
Tier-1 determinism	Archive rows	282	—	172 exact
Cross-host re-run	History entries	200 evals	—	17 exact
Ritz vs. Euler column	$\lvert\Delta P_{cr}\rvert/P_{cr}$	—	—	5.0e-9

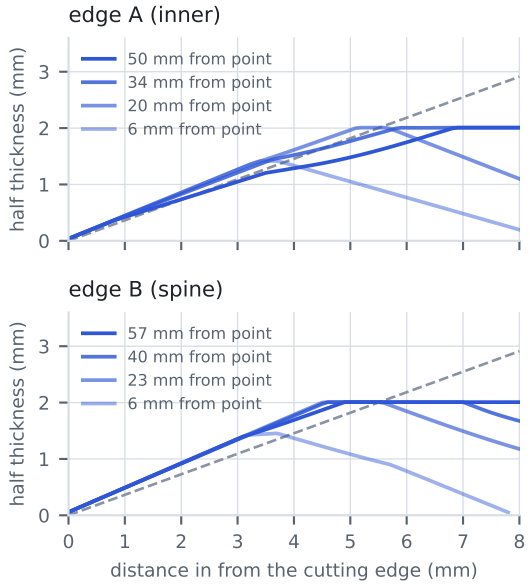


Figure 4 | Upper half of the blade cross-section at four stations along each edge of `kv4i1404_g00581_29ea3c73`, true scale, measured normal to the serrated edge from the exact decoder law (dashed: 20° per side, the requirement's minimum).

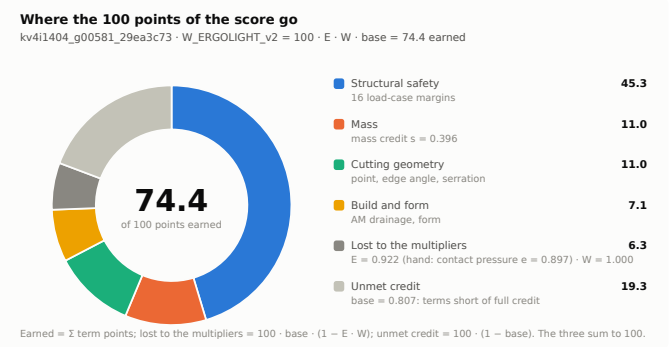


Figure 5 | The 100 points of the score of `kv4i1404_g00581_29ea3c73`: earned by group, lost to the multipliers ($100\,\text{base}\,(1 - E\,W)$) and unmet credit ($100\,(1 - \text{base})$). The three sum to 100 (g5-release fitness).

- **Blade surface texture (frozen G5.0 decoder limit).** The frozen G5.0 decoder allows the evolved surface texture to reach the blade zone, where its depth is tex_amp × scope_blade (for `kv4i1404_g00581_29ea3c73`, 0.22 × 0.066 = 0.014 mm; pitch 2.51 mm), so the as-decoded blade and its bevels are nearly smooth. The edge cross-sections (Fig. 4) and edge measurements evaluate the exact bevel law, which post-build grinding restores because the RS-karambit-v5 finishing allowance (0.5 mm per ground face) exceeds the texture depth. Keeping texture off the bevels by construction is deferred to a later decoder generation.
- **One platform of record.** v5.0.0's arm64 records agreed closely with, but were not bit-identical to, x86_64 re-evaluations, because the platforms' maths libraries differ in the last bits. v6.0.0 runs the search, every re-evaluation and the ranking on one platform (static x86_64 binaries, natively or in a linux/amd64 container): every archive row re-scores bit for bit, every pool is recomputed from its sources, `epoch01/xbabe4/island101` re-run on xbabe2 reproduced all 17 logged history entries over 200 evaluations, and re-evaluating all 282 elites of the final pool reproduced all 172 Tier-1 rows. A different libm (another platform) would again agree closely rather than exactly.
- **The search grid can rank designs it cannot keep.** The search scores at $h = 0.4$ mm with a section model of the tip; the ranking re-scores the shortlist at $h = 0.2$ mm with the solid tip and the winner must also pass the grid check and every load case on every plate mesh. A design that leads the search grid can therefore rank lower, which is why the search-grid best of the lineage (75.54) is not the score of record (74.40).
- **Analytic hand-zone models, checked by FE.** The score models grip squeeze, guard push and the serration teeth

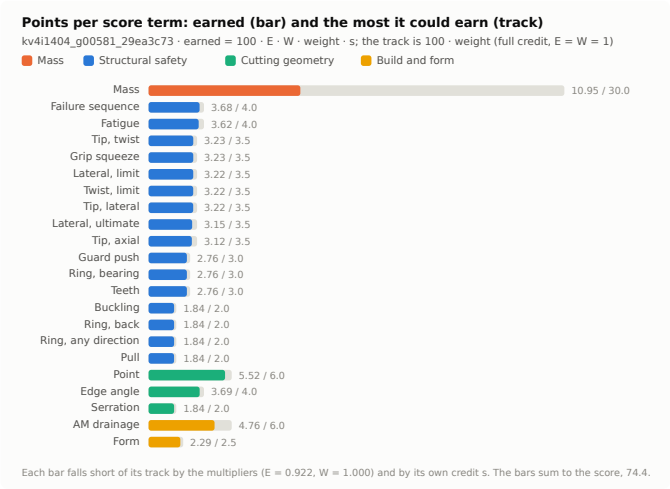


Figure 6 | Points earned by each weighted term of `kv4i1404_g00581_29ea3c73` (bars, coloured by group) against 100× its weight, the most it could earn at full credit with $E = W = 1$ (tracks).

Table 6 | Independent hex8 voxel FE against the score's analytic models (`kv4i1404_g00581_29ea3c73`; safety factor, required 1.20)

Case	Score model	Hex8 FE
Grip squeeze (800 N)	2.32	23.78
Guard push (300 N)	26.39	8.85
Serration teeth (180/60 N)	2.38	2.72

analytically. Against an independent hex8 voxel FE of the same design (Table 6), the score's safety factor is below the FE for grip squeeze (2.32 against 23.78), above it for guard push (26.39 against 8.85) and below it for the teeth (2.38 against 2.72); a model above the FE is optimistic. The lowest FE safety factor, 2.72, clears the 1.20 requirement. The FE checks are reported, not scored, so a later decoder could score these cases on the solid.

- **Linear-elastic surrogates and solid-tip readout windows.** All structural evaluations are linear-elastic. Permanent set under the GOST R 51501-99 deflection test [62] is represented by no yield at SF = 1.20, not by an elastoplastic solve. The 3D TET10 tip submodel reports SF = $\sigma_{\text{allow}}/\sigma_{\text{vM,max}}$ from the nodal maximum von Mises stress within each case's readout window ($r \in [1.0, 10.0]$ mm for axial, $[1.0, 2.5]$ mm for side load, and $[4.5, 15.0]$ mm for twist) to stay clear of the loaded patch (0.75 mm axial/lateral, 1.0 mm twist) and the clamped cut plane ($r_{\text{cut}} = 30$ mm).
- **Homogenised TPMS core and first-order hand patch.** In the blade and grip section models, the sheet-TPMS lattice is represented by the Naghavi *et al.* [31] and Timercan *et al.* [32] compression scaling laws; periodic unit-cell finite-element homogenisation and tension-fatigue testing of the printed sheet lattice are not yet performed. Hand contact pressure uses a first-order patch area (60 % of grip length times half the swell perimeter) rather than a deformable soft-tissue contact solve.
- **Evidence gaps requiring physical coupon and part tests (PROVENANCE-v5.md).** As documented in PROVENANCE-v5.md, ten domain inputs rest on derived bounds or single-orientation studies and require physical test replacement before any fitness-for-use claim: per-tooth snag/pry loads (180/60 N), off-axis tip side/twist shares (10 N, 0.30 N·m), ISO 8442-5 [61] edge retention of ground L-PBF Ti-6Al-4V, little-finger ring avulsion hazard [41], wet skin-on-titanium friction ($\mu = 0.15$), down-skin thin-wall roughness depth ($T_s = 0.10$ mm), 80 °C L-PBF tensile derating, and the 1.22 m MIL-STD-810H transit drop test [64], [65].

Table 7 | In-git CAD meshes of `kv4i1404_g00581_29ea3c73` (files <design>_<name>) and release assets (files g5-v6.0.0-[<design>_]<name>)

File	Triangles	MiB	Where
`AMbuild_mm.stl`	1,030,568	49.14	git (cad/)
`envelope_mm.stl`	925,952	44.15	git (cad/)
`archives.tar.gz`	—	167.16	release
`eval_h040.jsonl`	—	1.29	release
`AMbuild_h0.2_mm.stl`	—	361.28	release
`envelope_h0.2_mm.stl`	—	116.95	release

11 Reproducibility

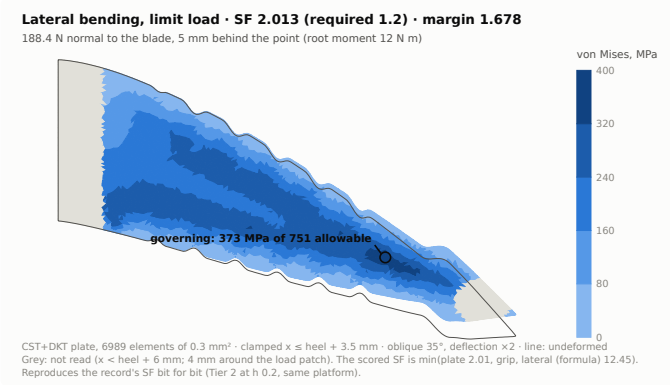
The cohort of record v6.0.0 ran the frozen decoder (hash `c63629d4b310`, freeze label `bf3e37fa`). The winner's 134-gene row is committed at evidence/g5/v6.0.0/winner_genome.csv, beside the single release receipt evidence/g5/v6.0.0/release.json, every island's launch record and history, every pool's recorded sources, the final pool, the Tier-2 ranking, the grid check, the plate-mesh check, the cross-host re-run, the v5.0.0 winner's records on the same platform, and the GPU-spike and fleet-sizing measurements. g5-release receipt-v6 writes the receipt only after recomputing every merge of the lineage byte for byte and checking every link (Section 6.3); g5-release-tex generates every macro in docs/g5/paper/generated/, and ops/ci/gen_readme.py generates the root README.md and cad/README.md from the same receipt. An island is reproduced by building the static binaries from this release and running g5-continue run with the arguments of its launch record (--pool, --evals, --seed, --host, --island, --commit `bf3e37fa`, --h 0.4); a pool, with g5-continue merge over its recorded sources in order. The in-git meshes and the release assets are listed in Table 7.

G5 leader renders (current pipeline)

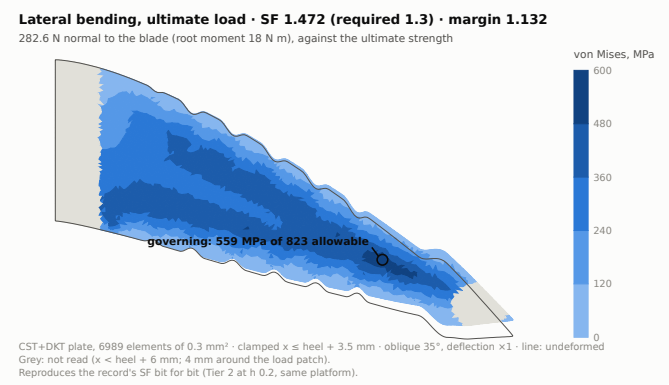
The current G5 pipeline's leader `kv4i1404_g00581_29ea3c73` (score 74.40, mass 88.4 g, internal void fraction 0.53) rendered in Blender from its AM-build STL. Cuts are open so the printed internal lattice is visible; finishes are illustrative.

12 Conclusion

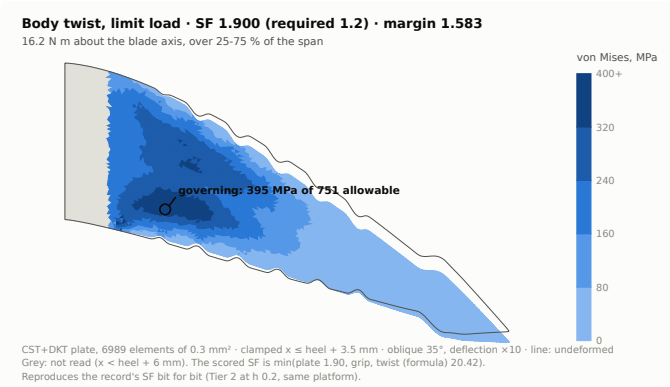
Rebuilding the Jarambit pipeline around the literature-sourced requirement set RS-karambit-v5 and the G5.0 decoder addressed the representation, load-case, drainage and objective flaws exposed by the G3 diagnostics. The grip perimeter, palm-swell aspect ratio, rounded-gullet serrations, sheet-TPMS wall



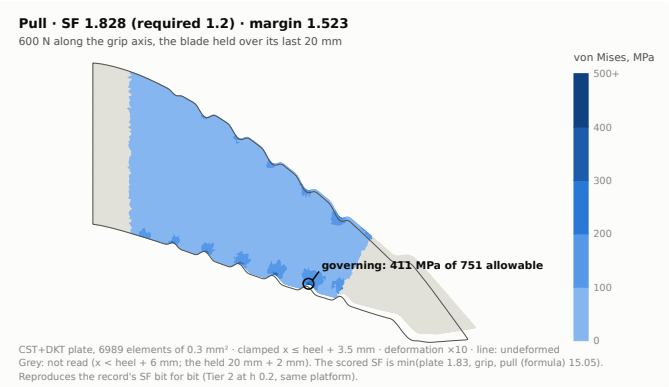
Lateral bending, limit load: SF 2.01



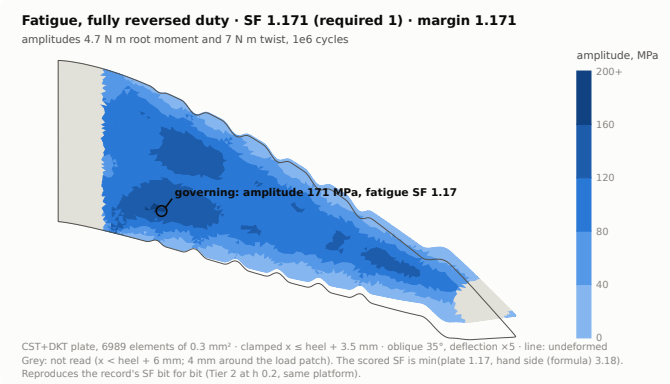
Lateral bending, ultimate load: SF 1.47



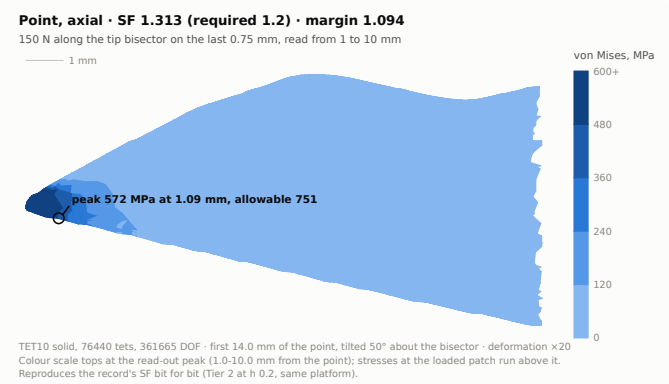
Body twist, limit load: SF 1.90



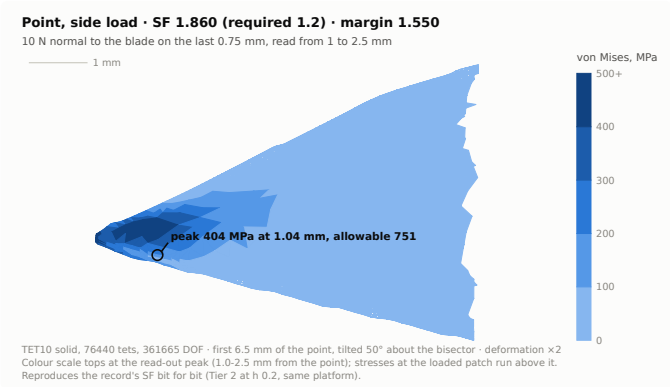
Pull: SF 1.83



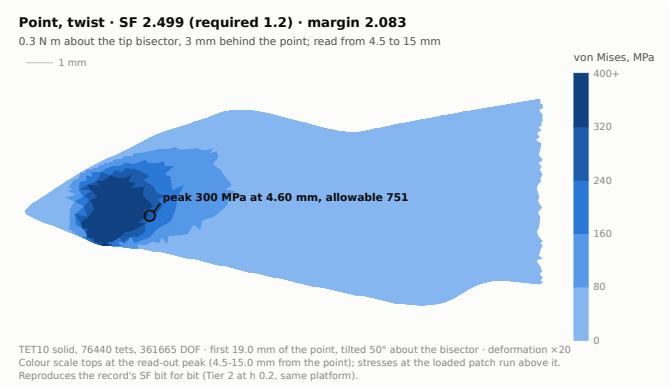
Fatigue, fully reversed duty: SF 1.17



Point, axial: SF 1.31

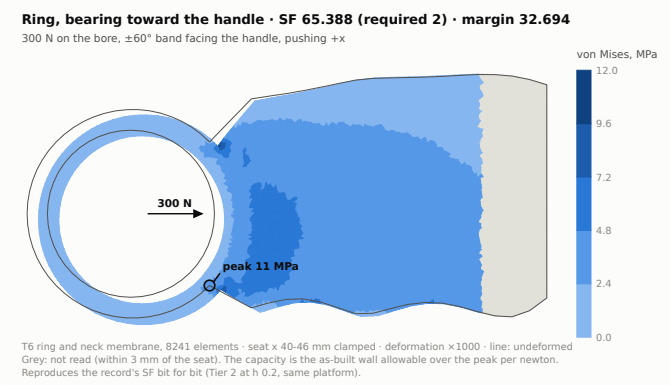


Point, side load: SF 1.86

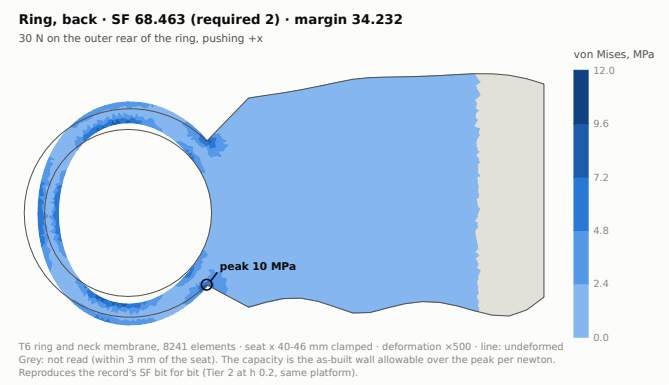


Point, twist: SF 2.50

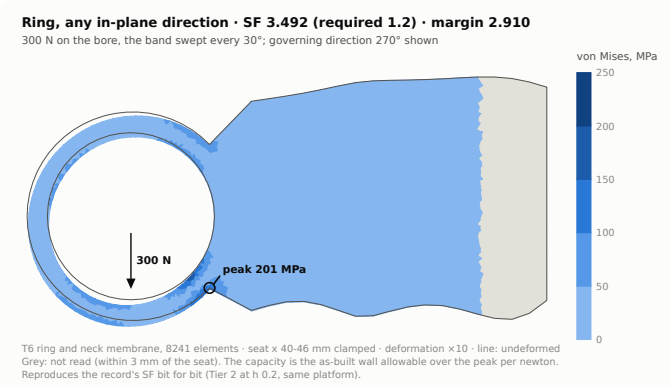
Figure 7 | FEM renderings of the load cases of `kv4i1404_g00581_29ea3c73` (1 of 2): von Mises stress, governing point and deformed shape, as re-solved by g5-fem-fields; each panel states its mesh, load and safety factor.



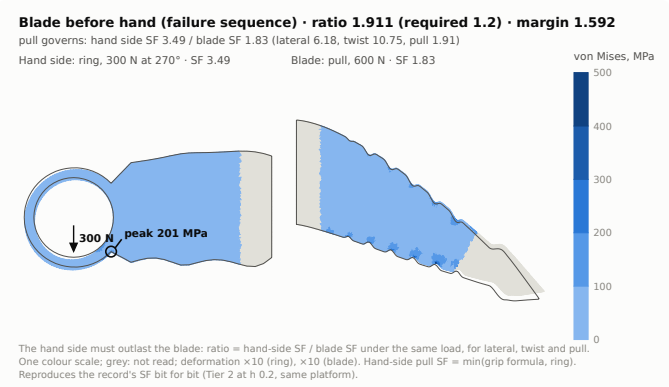
Ring, bearing toward the handle: SF 65.39



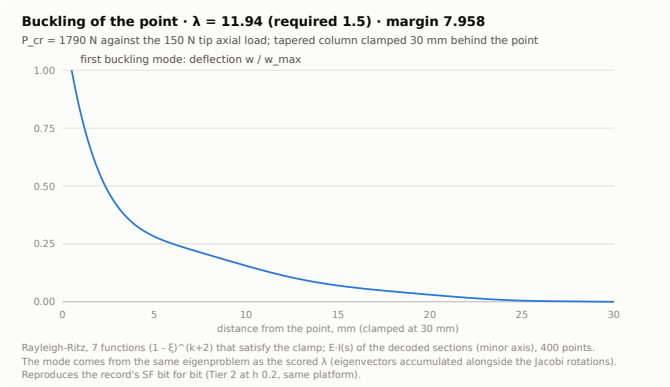
Ring, back: SF 68.46



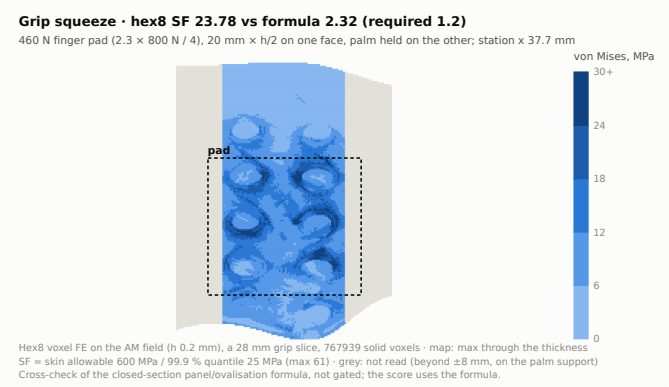
Ring, any in-plane direction: SF 3.49



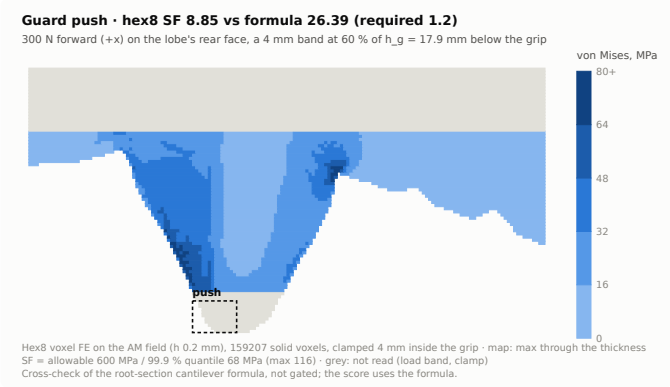
Blade before hand (failure sequence): SF 1.91



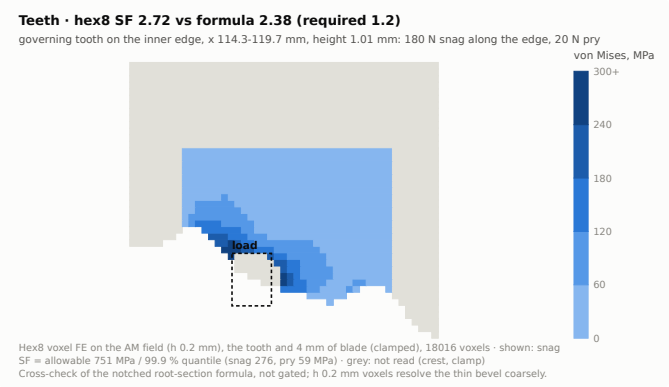
Buckling of the point: SF 11.94



Grip squeeze: hex8 FE SF 23.78 (score model 2.32)



Guard push: hex8 FE SF 8.85 (score model 26.39)



Teeth: hex8 FE SF 2.72 (score model 2.38)

Figure 8 | FEM renderings of the load cases of `kv4i1404_g00581_29ea3c73` (2 of 2), including the hex8 checks of the analytic cases.

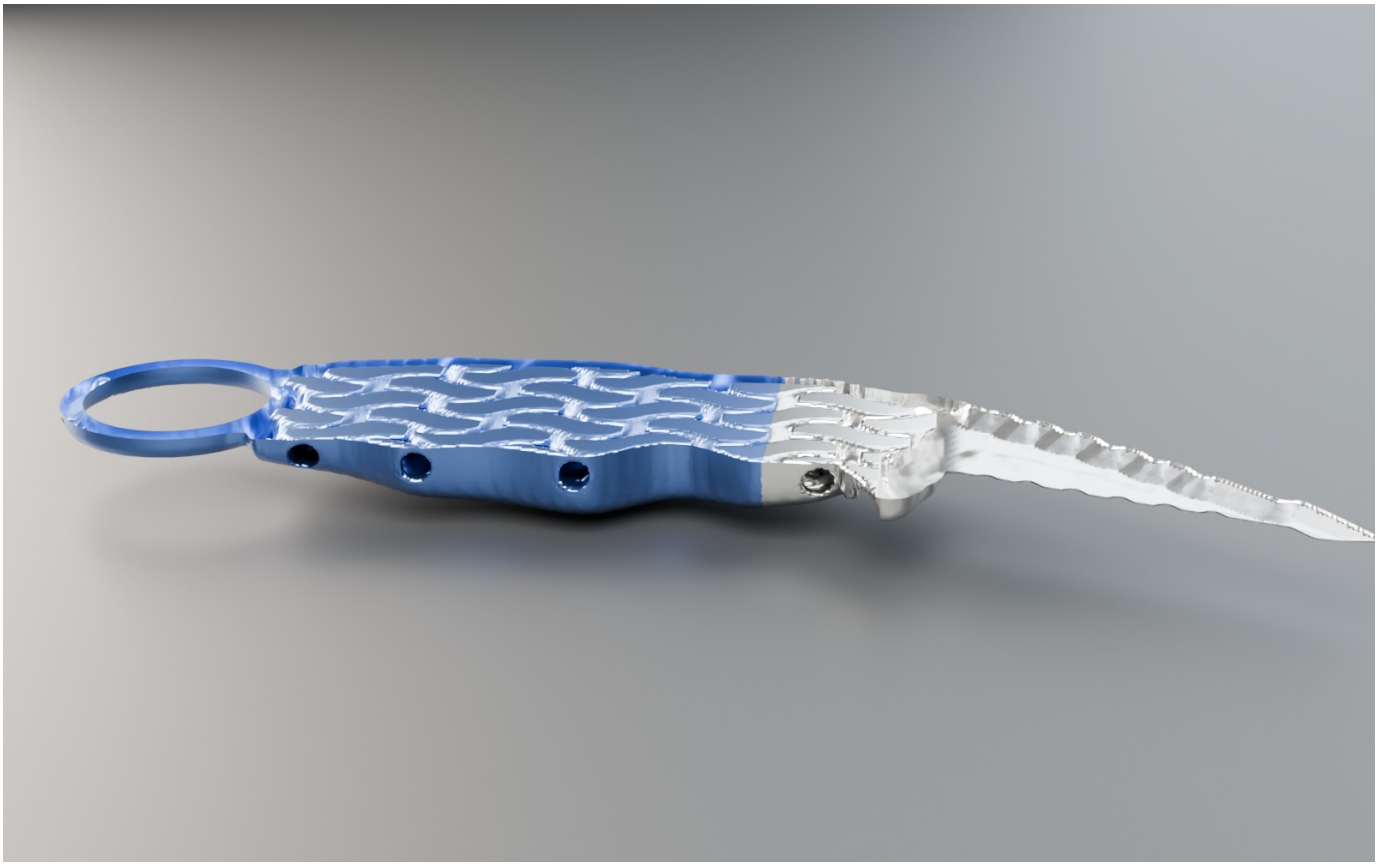


Figure 9 | X-ray cutaway of `kv4i1404_g00581_29ea3c73`: internal 3D-printed lattice voids.

thickness and rib-clear drainage ports are expressed directly in requirement units, so every candidate has a full-sized handle and width-checked drainage; the point angle is a continuous penalty rather than a construction rule (the winner measures 60.4° at 3 mm against a 60° target). In W_ERGOLIGHT_v2 the multiplicative worst-margin factor W and the hand multiplier E (the 5.0th power of the weighted product of the hand factors) make it costly, not impossible, to trade structural adequacy or hand comfort for grams. v6.0.0 continued the v5.0.0 cohort on one platform of record: from the re-evaluated v5.0.0 elites, 247 warm-started islands in 20 rounds (296,400 evaluations) raised the best search-grid score from 63.36 to 75.54, above the best of 4 fresh-start control islands (59.56). The best individual `kv4i1404_g00581_29ea3c73` scores 74.40 at Tier 2 (h = 0.2 mm), higher than the v5.0.0 winner on the same platform (59.34), weighs 88.41 g (finished partial volume; 90.29 g print mesh) and meets every load case, with worst margin ratio $\mu = 1.094$ as the minimum over the plate meshes 0.6/0.3/0.15 mm^2 and hand multiplier $E = 0.922$. Of its 100 points it earns 74.4; the multipliers take 6.3 and 19.3 is unmet credit. It is a computational design: printing, stress relief, grinding of the bevels and mechanical tests against the evidence gaps of Section 10 are the next steps, before any claim of fitness for use.

References

[1] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," *arXiv:1504.04909*, 2015.

[2] V. Vassiliades, K. Chatzilygeroudis, and J.-B. Mouret, "Using centroidal Voronoi tessellations to scale up the multidimensional archive of phenotypic elites algorithm," *IEEE Trans. Evol. Comput.*, vol. 22, no. 4, pp. 623–630, 2018.

[3] V. Vassiliades and J.-B. Mouret, "Discovering the elite hypervolume by leveraging interspecies correlation," in *Proc. GECCO*, 2018.

[4] S. P. Lloyd, "Least squares quantization in PCM," *IEEE Trans. Inf. Theory*, vol. 28, no. 2, pp. 129–137, 1982, doi:10.1109/TIT.1982.1056489.

[5] S. Joe and F. Y. Kuo, "Constructing Sobol sequences with better two-dimensional projections," *SIAM J. Sci. Comput.*, vol. 30, no. 5, pp. 2635–2654, 2008.

[6] J. Matoušek, "On the L_2-discrepancy for anchored boxes," *J. Complexity*, vol. 14, no. 4, pp. 527–556, 1998.

[7] A. Doi and A. Koide, "An efficient method of triangulating equi-valued surfaces by using tetrahedral cells," *IEICE Trans. Inf. Syst.*, vol. E74-D, no. 1, pp. 214–224, 1991.

[8] W. E. Lorensen and H. E. Cline, "Marching cubes: A high resolution 3D surface construction algorithm," *Comput. Graph.*, vol. 21, no. 4, pp. 163–169, 1987.

[9] J.-L. Batoz, K.-J. Bathe, and L.-W. Ho, "A study of three-node triangular plate bending elements," *Int. J. Numer. Methods Eng.*, vol. 15, no. 12, pp. 1771–1812, 1980, doi:10.1002/nme.1620151205.

[10] P. J. Roache, *Verification and Validation in Computational Science and Engineering*. Albuquerque, NM: Hermosa Publishers, 1998.

[11] C. E. Inglis, "Stresses in a plate due to the presence of cracks and sharp corners," *Trans. Inst. Naval Archit.*, vol. 55, pp. 219–241, 1913.

[12] S. P. Timoshenko and J. M. Gere, *Theory of Elastic Stability*, 2nd ed. New York: McGraw-Hill, 1961.

[13] ASTM F2924-14(2021), *Standard Specification for Additive Manufacturing Titanium-6 Aluminum-4 Vanadium with Powder Bed Fusion*, ASTM International, 2021.

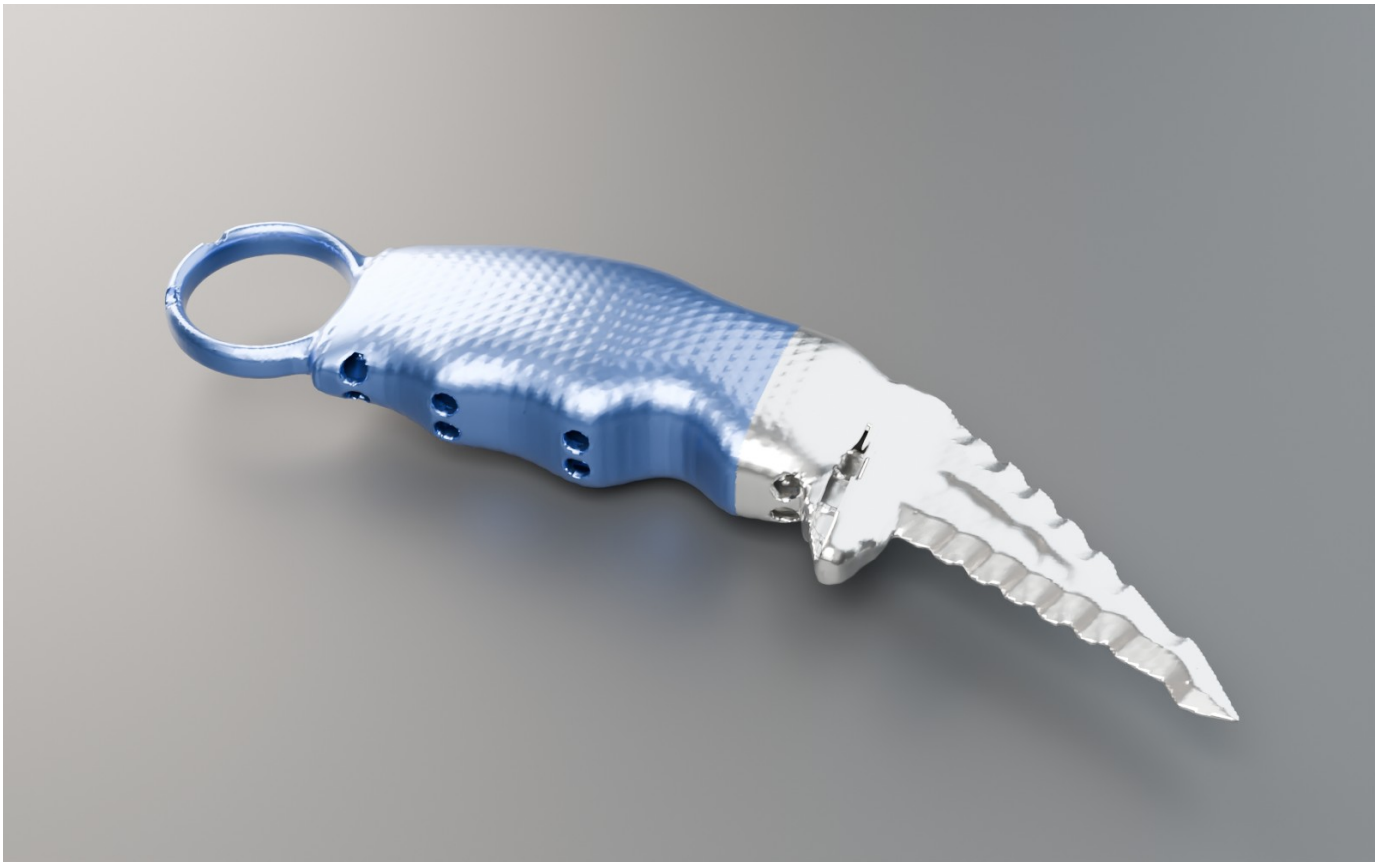


Figure 10 | G5 leader `kv4i1404_g00581_29ea3c73`, oblique.

[14] U.S. Department of Defense, *MIL-HDBK-5J: Metallic Materials and Elements for Aerospace Vehicle Structures*, §5.4.1 (Ti-6Al-4V annealed: Table 5.4.1.0(b); Figs. 5.4.1.1.1 and 5.4.1.1.4), 2003.

[15] NIAR/NCAMP, *ND-ME 004: L-PBF Ti-6Al-4V Material Specification*, 2025.

[16] NASA, *NASA-STD-6030: Additive Manufacturing Requirements for Spaceflight Systems*, 2021.

[17] EOS GmbH, *EOS Titanium Ti64 Grade 5 Material Data Sheet (M290)*.

[18] B. Vrancken, L. Thijs, J.-P. Kruth, and J. Van Humbeeck, "Heat treatment of Ti6Al4V produced by selective laser melting," *J. Alloys Compd.*, vol. 541, p. 177, 2012, doi:10.1016/j.jallcom.2012.07.022.

[19] V. Cain *et al.*, "Crack propagation and fracture toughness of Ti6Al4V alloy produced by selective laser melting," *Addit. Manuf.*, vol. 5, p. 68, 2015, doi:10.1016/j.addma.2014.12.006.

[20] Y. Xie *et al.*, *Materials*, vol. 19, art. 1945, 2026, doi:10.3390/ma19101945.

[21] S. Murchio *et al.*, *Int. J. Fatigue*, vol. 180, art. 108102, 2024, doi:10.1016/j.ijfatigue.2023.108102.

[22] M. Benedetti *et al.*, *Int. J. Fatigue*, vol. 107, p. 96, 2018, doi:10.1016/j.ijfatigue.2017.10.021.

[23] G. Nicoletto, *Int. J. Fatigue*, vol. 106, p. 124, 2018, doi:10.1016/j.ijfatigue.2017.10.004.

[24] E. Wycisk *et al.*, *Phys. Procedia*, vol. 56, p. 371, 2014, doi:10.1016/j.phpro.2014.08.120.

[25] E. Wycisk *et al.*, *Front. Mater.*, vol. 2, art. 72, 2015, doi:10.3389/fmats.2015.00072.

[26] L. Pagliari *et al.*, *Eng. Fail. Anal.*, vol. 180, art. 109825, 2025, doi:10.1016/j.engfailanal.2025.109825.

[27] J. Johnsen *et al.*, *Welding in the World*, vol. 68, p. 361, 2024, doi:10.1007/s40194-023-01604-5.

[28] M. Kahlin, H. Ansell, and J. Moverare, *Int. J. Fatigue*, vol. 101, p. 51, 2017, doi:10.1016/j.ijfatigue.2017.04.009.

[29] S. Beretta and S. Romano, *Int. J. Fatigue*, vol. 94, p. 178, 2017, doi:10.1016/j.ijfatigue.2016.06.020.

[30] L. Barricelli, L. Patriarca, A. du Plessis, and S. Beretta, *Int. J. Fatigue*, art. 107401, 2022, doi:10.1016/j.ijfatigue.2022.107401.

[31] S. A. Naghavi *et al.*, *Bioengineering*, vol. 9, art. 504, 2022, doi:10.3390/bioengineering9100504.

[32] A. Timercan, V. Sheremetyev, and V. Brailovski, *Sci. Technol. Adv. Mater.*, vol. 22, p. 285, 2021, doi:10.1080/14686996.2021.1907222.

[33] C. N. Kelly *et al.*, *Acta Biomater.*, vol. 94, p. 610, 2019, doi:10.1016/j.actbio.2019.05.046.

[34] K. Lietaert *et al.*, *Sci. Rep.*, vol. 8, art. 4957, 2018, doi:10.1038/s41598-018-23414-2.

[35] M. Benedetti, A. du Plessis *et al.*, *Mater. Sci. Eng. R*, vol. 144, art. 100606, 2021, doi:10.1016/j.mser.2021.100606.

[36] M. Alaña *et al.*, *Sci. Rep.*, vol. 11, 2021, doi:10.1038/s41598-021-98631-3.

[37] Renishaw, *Design for Metal AM: A Beginner's Guide*, 2017.

[38] A. Carrozza, B. Bircher, A. Aversa, and S. Biamino, "Investigating complex geometrical features in LPBF-produced parts: a material-based comparison between different titanium alloys," *Met. Mater. Int.*, 2023, doi:10.1007/s12540-023-01460-4.

[39] Xometry DMLS design guide; Materialise metal design guide; Solukon depowdering guidance (powder ports 2–5 mm, ≥ 3 mm).

[40] R. M. Dodds *et al.*, "Grip strength across the life course," *PLoS ONE*, vol. 9, art. e113637, 2014, doi:10.1371/journal.pone.0113637.

[41] NASA, *Human Integration Design Handbook (HIDH)*, NASA/SP-2010-3407/REV1, §9.12.2 and App. B Table 8, 2014.

[42] U.S. Department of Defense, *MIL-STD-1472H: Human Engineering*, §5.7.6.3–4 and Fig. 23, 2020.

[43] Cal/OSHA–NIOSH, *Easy Ergonomics: A Guide to Selecting Non-Powered Hand Tools*, Pub. 2004-164, 2004.

[44] Y.-K. Kong and B. D. Lowe, "Optimal cylindrical handle diameter for grip force tasks," *Int. J. Ind. Ergon.*, vol. 35, p. 495, 2005, doi:10.1016/j.ergon.2004.11.003.

[45] N. J. Seo and T. J. Armstrong, "Investigation of grip force, normal force, contact area, hand size, and handle size for cylindri-

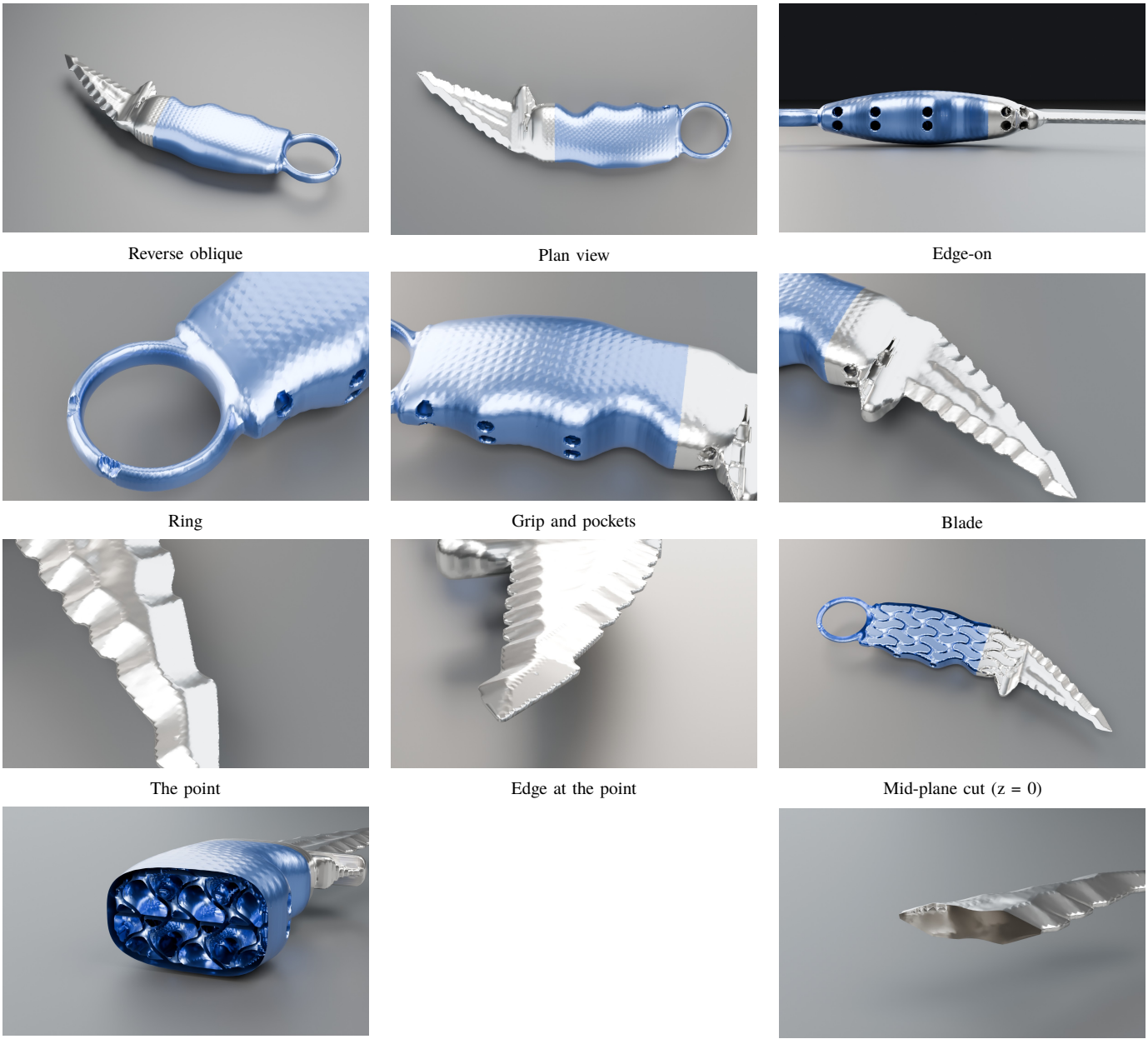


Figure 11 | All Blender views of the G5 leader.

cal handles," *Human Factors*, vol. 50, no. 5, pp. 734–744, 2008, doi:10.1518/001872008X354192.

[46] N. J. Seo and T. J. Armstrong, "Effect of elliptic handle shape on … twisting ability," *Ergonomics*, vol. 54, p. 961, 2011, doi:10.1080/00140139.2011.606923.

[47] C. C. Gordon *et al.*, *ANSUR II*, NATICK/TR-15/007, 2014.

[48] Federal Aviation Administration, *Human Factors Design Standard (HFDS)*, App. B (Anthropometry).

[49] S. F. Tirloni *et al.*, "Cutting force measurement …: a systematic review," *EXCLI J.*, vol. 20, p. 727, 2021, doi:10.17179/excli2021-3167.

[50] D. J. Cochran and M. W. Riley, "An evaluation of knife handle guarding," *Human Factors*, vol. 28, p. 295, 1986, doi:10.1177/001872088602800305.

[51] B. Buchholz, L. J. Frederick, and T. J. Armstrong, *Ergonomics*, vol. 31, p. 317, 1988, doi:10.1080/00140138808966676.

[52] C. Fransson-Hall and Å. Kilbom, "Sensitivity of the hand to surface pressure," *Appl. Ergon.*, vol. 24, p. 181, 1993.

[53] L. Johansson *et al.*, "Perception of surface pressure applied to the hand," *Ergonomics*, vol. 42, p. 1274, 1999, doi:10.1080/001401399184947.

[54] Y. Aldien *et al.*, *Int. J. Ind. Ergon.*, vol. 35, p. 267, 2005, doi:10.1016/j.ergon.2004.09.005.

[55] L. W. O'Sullivan and T. J. Gallwey, *J. Electromyogr. Kinesiol.*, vol. 12, p. 275, 2002, doi:10.1016/S1050-6411(02)00014-7.

[56] Y. Yoshii *et al.*, *BioMed. Eng. OnLine*, vol. 14, art. 115, 2015.

[57] Y.-K. Kong *et al.*, *Ergonomics*, 2007 (screwdriving torque).

[58] R. W. McGorry, P. C. Dowd, and P. G. Dempsey, *Appl. Ergon.*, 2005 (moments in knife use).

[59] T. J. Walilko, D. C. Viano, and C. A. Bir, "Biomechanics of the head for Olympic boxer punches to the face," *Br. J. Sports Med.*, vol. 39, no. 10, pp. 710–719, 2005.

[60] EN/ISO 8442-1 and -2, handle twist tests (via CATRA).

[61] ISO 8442-5:2004, *Materials and Articles in Contact with Foodstuffs—Cutlery and Table Holloware—Part 5: Specification for Sharpness and Edge Retention Test of Cutlery*, 2004.

[62] GOST R 51501-99, tourist knives, Clause 9.5.1, 1999.

[63] U.S. Department of Defense, *MIL-K-8662E: Pilot Survival Knife*, 1974.

[64] U.S. Department of Defense, *MIL-STD-810H*, Method 516.8, Procedure IV.

[65] ANSI/PLATO FL1-2019 (drop ≥ 1 m onto concrete).
[66] EN 388 puncture levels (20 / 60 / 100 / 150 N).
[67] M. D. Gilchrist *et al.*, *Forensic Sci. Int.*, vol. 177, p. 52, 2008.
[68] G. Nolan, S. V. Hainsworth, and G. N. Rutty, *J. Forensic Sci.*, vol. 58, p. 372, 2013.
[69] S. V. Hainsworth *et al.*, *Int. J. Legal Med.*, vol. 122, p. 281, 2008, doi:10.1007/s00414-007-0202-6.
[70] L. Thomas, `knifesteelnerds.com`, CATRA regression and edge-geometry articles, 2018.
[71] Y. Zhang *et al.*, *Materials*, vol. 16, art. 5375, 2023.
[72] R. Landes, *Messerklingen und Stahl*, 2006 (via Thomas).
[73] U.S. Patent 4,823,467 (serrated knife).